

Brainstem

The brainstem consists of the medulla oblongata, pons and midbrain. It is situated in the posterior cranial fossa, and its ventral surface lies on the clivus. It contains numerous intrinsic neuronal cell bodies and their processes, some of which are the brainstem homologues of spinal cell groups. Some brainstem cell groups are the nuclei of cranial nerves and are concerned with the sensory, motor and autonomic innervation of the head and neck. Other autonomic (visceromotor) fibres that arise from the brainstem are distributed widely within the thorax and abdomen via branches of the vagus nerve. The brainstem also contains a complex and sometimes ill-defined network of neurones, known as the reticular formation (Haines 2013, Olszewski and Baxter 1954), which extends throughout its length, and is continuous caudally with its spinal counterpart. Some reticular nuclei are referred to as vital centres because they are concerned with regulation of cardiac and respiratory activities; other parts of the reticular formation are essential for cerebral cortical arousal and the maintenance of consciousness, or are involved in the regulation of muscle tone, posture and reflex activities. The brainstem is the site of origin and/or termination of numerous ascending and descending fibres and is traversed by many others. The anterolateral system (spinal lemniscus), medial lemniscus and the trigeminothalamic tracts all ascend through the brainstem to reach the thalamus. Prominent corticospinal projections descend through the brainstem and corticonuclear projections end within it.

Damage to the brainstem is often devastating and life-threatening (Posner et al 2007). This is because it is a structurally and functionally compact region, where even small lesions can destroy vital centres (cardiac and respiratory), disconnect forebrain motor areas from brainstem and spinal motor neurones, and sever incoming sensory fibres from higher centres of consciousness, perception and cognition. Irreversible cardiac and respiratory arrest follows complete destruction of the neural respiratory and cardiac centres in the medulla. Clinically, this is part of a complex event called brain death (cerebral death), a condition that includes cessation of brainstem function. Accurate diagnosis is essential because brainstem death may occur in patients whose respiratory and cardiac functions may be artificially maintained indefinitely on life support.

THE BRAINSTEM IN A CLINICAL CONTEXT

The advent of computed tomography (CT) and magnetic resonance imaging (MRI) has revolutionized clinical medicine and forever altered how we look at the brainstem in an educational setting. This is especially the case when the goal is to teach neuroanatomical concepts in a manner most applicable to the way in which these concepts are viewed and used in the clinical environment.

The time-honoured approach to teaching brainstem neurobiology is in an anatomical orientation, where dorsal/posterior is 'up' in the image and ventral/anterior is 'down' (Figs 21.1–21.2). Location of internal structures and the arrangement of somatotopic maps follow accordingly. Learning neuroanatomy in this orientation is correct but may not be the most appropriate for the clinical setting. Moreover, there is no standard of laterality in brainstem images viewed anatomically; the observer's right may be either the right or the left side. These inconsistencies disappear when the brainstem is taught using a clinical orientation: the shape of the image (stained section or line drawing) matches that of the corresponding MRI or CT (see Figs 21.1–21.2); the orientation of all images, irrespective of their source (stained section, drawing, CT, MRI), is the same, i.e. the observer's right is the patient's left and the observer's left is the patient's right; and so the concept of laterality, which is enormously important in diagnosis, is reinforced for the student. When learned in a clinical orientation, somatotopically organized tracts and pathways map exactly on to their counterparts in a patient (see Fig. 21.1). In recognition of the inherent value of contextualizing neuroanatomical concepts within clinical reality, examples will be integrated into this chapter.

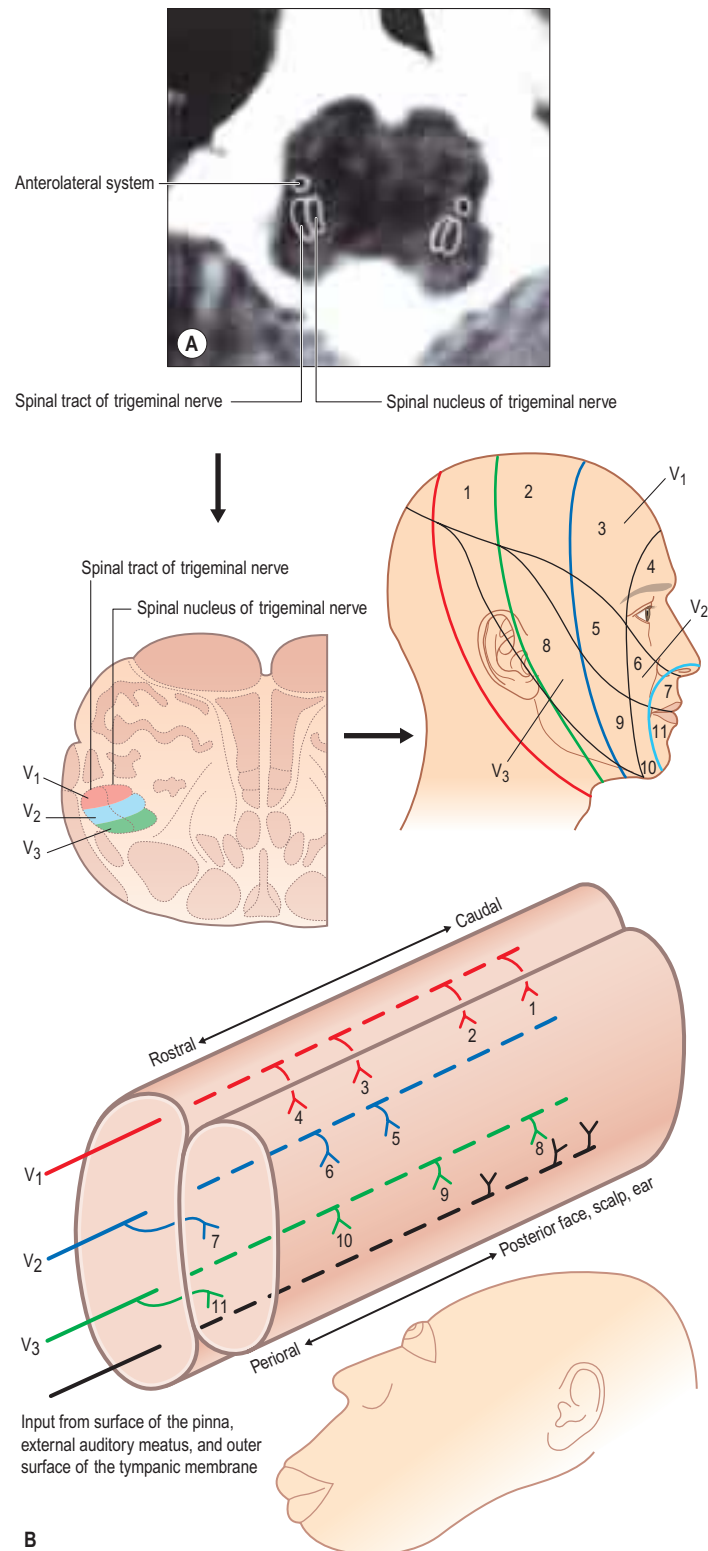


Fig. 21.1 Somatotopy of the spinal tract of the trigeminal nerve and its termination in the spinal nucleus, mainly the pars caudalis, in clinical orientation. Note the dorsal–ventral arrangement of the tract and the rostral–caudal distribution of its terminals in the nucleus as correlated with the facial dermatomes. (Redrawn with permission from Haines D, *Fundamental Neuroscience for Basic and Clinical Applications*, 4th Edition, 2013, Saunders, Elsevier.)



Fig. 21.2 The topographic relationship of the midbrain regions and the adjacent cisterns, with comparable (A) T2- and (B) T1-weighted MR images. (Redrawn from Haines D, *Fundamental Neuroscience for Basic and Clinical Applications*, 4th Edition, 2013, Saunders, Elsevier.)

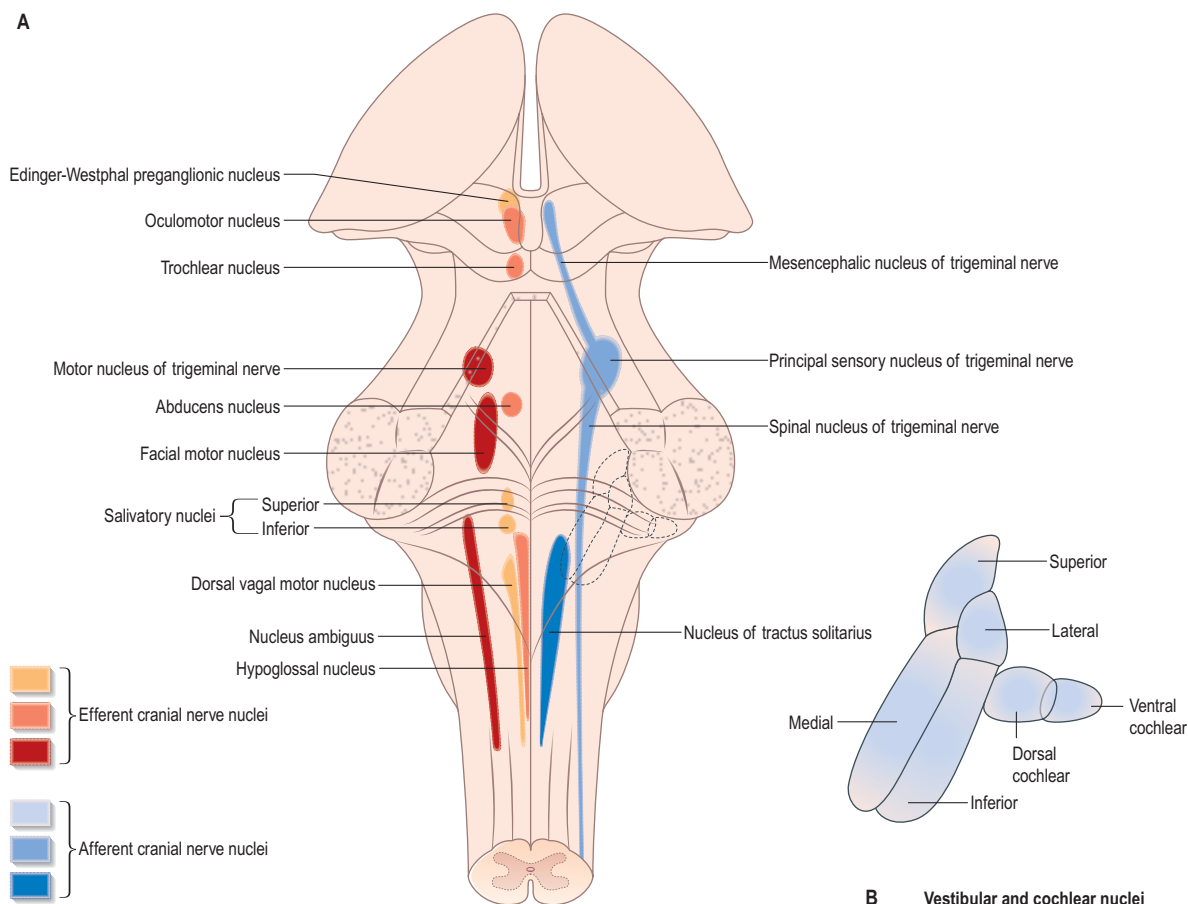


Fig. 21.3 A, The cranial nerve nuclei. The components of the vestibular nuclei are indicated by the dotted lines. **B**, The components of the vestibular and cochlear nuclei shown in more detail.



OVERVIEW OF CRANIAL NERVES AND CRANIAL NERVE NUCLEI

The cranial nerves are the routes by which the brain receives information directly from, and controls the functions of, structures that are located mainly, although not exclusively, within the head and neck. All but 3 of the 12 pairs of cranial nerves are regarded as being attached to the brainstem, the exceptions being the olfactory, optic and accessory nerves.

The cranial nerves are individually named and numbered, using Roman numerals, in a rostro-caudal sequence, reflecting their order of attachment to the brain (Haines 2015) (see Table 16.1). The olfactory nerve (I) arises from the olfactory epithelium in the nasal cavity and terminates directly in cortical and subcortical areas of the frontal and temporal lobes; it is closely associated functionally with the limbic system and is described in that context. The axons of the optic nerve (II) pass into the optic chiasma, where medially positioned axons decussate; all of the axons emerge as the optic tract, which terminates in

the lateral geniculate nucleus of the thalamus (the visual pathways are discussed in Ch. 42). The oculomotor (III) and trochlear (IV) nerves arise in, and are attached to, the midbrain. The trigeminal nerve (V) attaches to the pons, medial to the middle cerebellar peduncle. The abducens (VI), facial (VII) and vestibulocochlear (VIII) nerves attach to the brainstem at, or close to, the pontomedullary junction. The glossopharyngeal (IX), vagus (X) and hypoglossal (XII) nerves all attach to the medulla. The accessory nerve (XI) arises from cervical levels C1–C5/C6, enters the cranium through the foramen magnum and then exits via the jugular foramen, together with the glossopharyngeal and vagus nerves.

Cranial nerves III–X and XII are associated with brainstem cell groupings referred to collectively as the cranial nerve nuclei (Fig. 21.3). The latter are either the origin of efferent cranial nerve fibres or the site of termination of cranial nerve or descending cerebral cortical afferents. They are considered to be organized into six discontinuous longitudinal cell columns corresponding to columns that may be identified in the embryo (see Fig. 16.1). Three columns are 'sensory' and three are 'motor' in function.

For well over a century, the accessory nerve has been described as having two roots: a cranial root composed of a variable number of small rootlets that emerge from the postolivary groove of the dorsolateral medulla, caudal to the rootlets of the vagus, and a spinal root composed of fibres derived from the cervical segments of the spinal cord. According to this description, the cranial rootlets carry axons of motor neurones located in the caudal part of the nucleus ambiguus, and the spinal rootlets carry axons of motor neurones located in a nucleus in the ventral horn of the cervical spinal cord in line with the nucleus ambiguus (Ch. 18). The spinal root ascends the vertebral canal and enters the skull via the foramen magnum. The cranial and spinal roots join, either before or within the jugular foramen, and the bundle so formed then divides into internal and external branches on leaving the jugular foramen. The internal branch joins the vagus nerve and its fibres innervate branchiomeric muscles via the pharyngeal and laryngeal branches of the vagus nerve. The external branch descends into the neck to innervate sternocleidomastoid and the upper and middle fibres of trapezius. In recent years, this description has been challenged: the cranial root has been redesignated as anatomically and functionally part of the vagus nerve, and the spinal root has become, *de facto*, the accessory nerve. The issue is not simply one of contentious anatomical terminology; rather, it reflects the need to clarify the relevant microsurgical

anatomy in order to minimize potential complications during surgery in the region. Paradoxically, meticulous cadaveric dissection studies, often undertaken stereomicroscopically, have failed to reach a consensus, finding either no connections between rootlets arising from the dorsolateral medulla and the spinal root of the accessory nerve because the cranial root was absent, or significant variations in the arrangement and size of plexiform connections between the cranial rootlets and the spinal root. Plexiform interconnections between neighbouring cranial nerves are not uncommon and are certainly not confined to the accessory and vagus nerves (Shoja et al 2014); their functional significance is often uncertain. (For further reading, see Krammer et al (1987), Lachman et al (2002), Ryan et al (2007), Wiles et al (2007), Benninger and McNeil (2010), Skinner (2011), Liu et al (2014), Tubbs et al (2014).)

In addition to the interaction between fibres of the corticospinal and rubrospinal systems at the spinal level, there is yet another factor that relates to the functional characteristics described above. Experimental lesions in primates that target either of these tracts are usually stereotactically placed, damaging very specific fibre populations. On the other hand, spontaneous lesions in humans invariably damage, in addition to corticospinal axons, other axon populations that directly, or indirectly, influence spinal alpha motor neurones. This is also a major contributor to the differential results seen in primates versus humans.

The sensory nucleus of the trigeminal nerve, which extends throughout the length of the brainstem and into the cervical spinal cord, represents a general somatic afferent cell column (Haines 2013, Haines 2015, Olszewski 1950). Its principal afferents are carried in the trigeminal nerve. General visceral afferents carried by the facial, glossopharyngeal and vagus nerves end in the nucleus solitarius of the medulla. The special somatic afferent column corresponds to the vestibular and cochlear nuclei, which are located beneath the vestibular area of the floor of the fourth ventricle.

The general somatic efferent cell column consists of four nuclei that lie near the midline and give rise to motor fibres that run in nerves of the same name. From rostral to caudal, these are the oculomotor, trochlear and abducens nuclei, which innervate the extraocular muscles, and the hypoglossal nucleus, which innervates all but one of the muscles of the tongue. The general visceral efferent, or parasympathetic, cell column is made up of the Edinger–Westphal preganglionic nucleus of the midbrain, the salivatory nuclei of the pons, and the vagal nucleus of the medulla. Cells in the special visceral efferent column innervate muscles derived from the branchial arches and lie in the trigeminal motor nucleus, the facial nucleus and the nucleus ambiguus.

MEDULLA OBLONGATA

EXTERNAL FEATURES AND RELATIONS

The medulla oblongata extends from just above the first pair of cervical spinal nerves to the lower border of the pons (Haines 2013). It is approximately 3 cm in length and 2 cm in diameter at its widest. The ventral surface of the medulla is separated from the basilar part of the occipital bone and apex of the dens by the meninges and occipito-axial ligaments. Caudally, the dorsal surface of the medulla is adjacent to the cisterna magna, a subarachnoid cistern between the cerebellar hemispheres.

The ventral and dorsal surfaces of the medulla (Figs 21.4–21.5) possess a longitudinal median fissure and sulcus, respectively, which are continuous with their spinal counterparts. Caudally, the ventral (anterior) median fissure is interrupted by the obliquely crossing fascicles of the motor decussation (pyramidal decussation). Rostrally, it ends at the pontine border in a diminutive depression, the foramen caecum. A prominent elongated ridge, the pyramid, which contains descending corticospinal axons, lies immediately lateral to the ventral median fissure. The lateral margin of the pyramid is indicated by a shallow ventrolateral sulcus. A linear series of rootlets forming the hypoglossal nerve emerge from this sulcus in line with the ventral spinal nerve roots. The abducens nerve emerges at the caudal border of the pons generally in line with the rootlets of the hypoglossal nerve. Caudally, most fibres of the pyramid taper, decussate and enter the lateral funiculus of the spinal cord. Lateral to the pyramid and the ventrolateral sulcus is an oval prominence, the olive (see Fig. 21.4), which contains the inferior olivary nuclear complex. The posterolateral sulcus is lateral to the olive; the glossopharyngeal and vagus nerves join the brainstem along the line of this sulcus, in line with the dorsal spinal nerve roots.

The spinal central canal extends into the caudal half of the medulla, migrating progressively more dorsally until it opens out into the lumen of the fourth ventricle at the obex. This divides the medulla into a closed part, which contains the central canal, and an open part, which contains the caudal half of the fourth ventricle (see Fig. 21.5).

In the closed part of the medulla, a shallow dorsal intermediate sulcus, on either side of the dorsal median sulcus, is continuous with its cervical spinal counterpart and indicates the location of the dorsal (posterior) columns (fasciculi gracilis and cuneatus). The ascending fasciculi are parallel to each other, but in the caudal medulla they diverge slightly to end in elongated swellings, the gracile and cuneate tubercles, produced by the subjacent nuclei gracilis and cuneatus respectively (see Fig. 21.5; Figs 21.6–21.7). Most fibres in the fasciculi synapse with neurones in their respective nuclei. The latter project to the contralateral thalamus, which, in turn, projects to the primary somesthetic cortex (see Fig. 20.10). The restiform body lies dorsolaterally in the medulla, forming a rounded ridge between the caudal part of the fourth ventricle and the glossopharyngeal and vagal rootlets on each side. It contains a wide variety of fibres (dorsal spinocerebellar, cuneocerebellar, olivocerebellar, reticulocerebellar, and other cerebellar afferents) (see Fig. 21.9). The restiform bodies on the two sides diverge and incline to enter the cerebellar hemispheres as the major component of the inferior cerebellar peduncles. They are crossed at this point by the striae medullares, which run medial to lateral on the ventricular floor (see Fig. 21.5). Usually, each inferior cerebellar peduncle also contains a

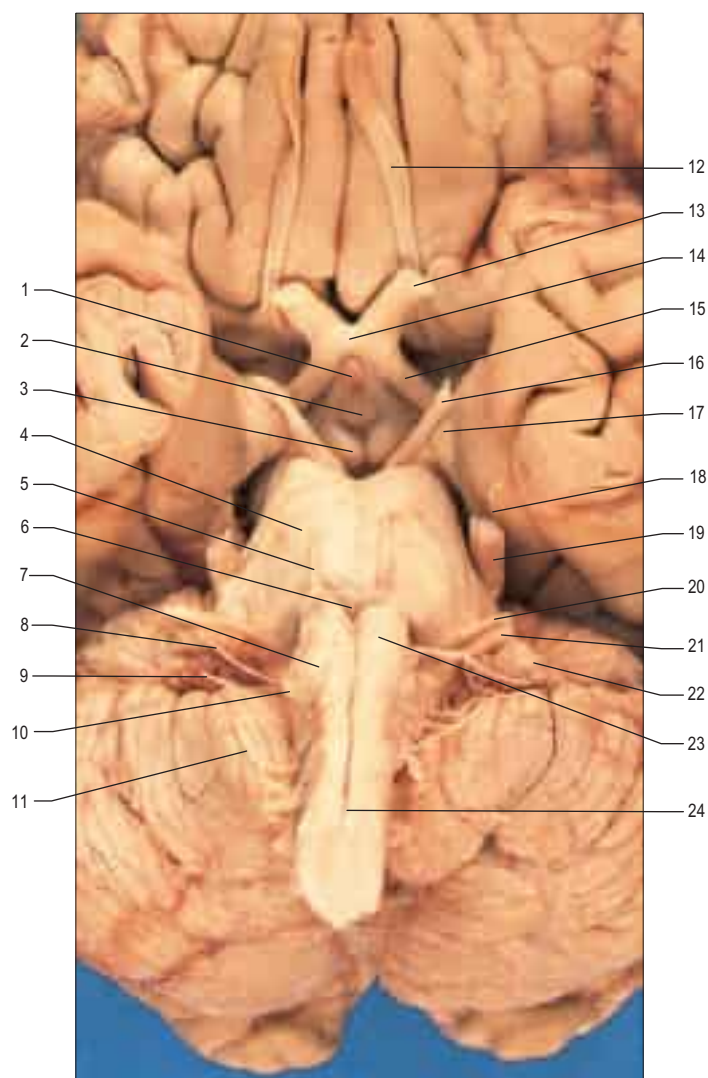


Fig. 21.4 The ventral aspect of the brainstem. Key: 1, infundibulum; 2, tuber cinereum; 3, mammillary body; 4, basilar pons; 5, abducens nerve; 6, foramen caecum; 7, olive; 8, glossopharyngeal nerve; 9, vagus nerve; 10, rootlets of hypoglossal nerve; 11, accessory nerve; 12, olfactory tract; 13, optic nerve; 14, optic chiasma; 15, optic tract; 16, oculomotor nerve; 17, uncus; 18, trochlear nerve; 19, trigeminal nerve; 20, facial nerve; 21, vestibulocochlear nerve; 22, flocculus; 23, pyramid; 24, motor decussation (decussation of pyramids).

small contribution from the juxtarestiform body (long Purkinje cell axons and efferents from the fastigial nuclei that traverse the vestibular nuclei; see Fig. 21.12). The inferior cerebellar peduncles form the anterior and rostral boundaries of the lateral recesses of the fourth ventricle; these are continuous with the subarachnoid space through the lateral apertures of the fourth ventricle (foramina of Luschka). A tuft of choroid plexus, continuous with that of the fourth ventricle, protrudes from the foramina on either side.

INTERNAL ORGANIZATION

Transverse section of the medulla at the level of the motor (pyramidal) decussation

A transverse section of the lower medulla oblongata (see Fig. 21.6) intersects the dorsal, lateral and ventral funiculi, which are continuous with their counterparts in the spinal cord. The ventral funiculi are separated from the central grey matter by corticospinal fibres, which cross in the motor (pyramidal) decussation to reach the contralateral lateral funiculi (see Fig. 21.16). The decussation displaces the central grey matter and central canal dorsally (Haines 2015). Continuity between the ventral grey column and central grey matter, which is maintained throughout the spinal cord, is lost. The column subdivides into the supraspinal nucleus (continuous above with that of the hypoglossal nerve), which is the efferent source of the first cervical nerve, and the nucleus of the accessory nerve, which is in line rostrally with the nucleus ambiguus.

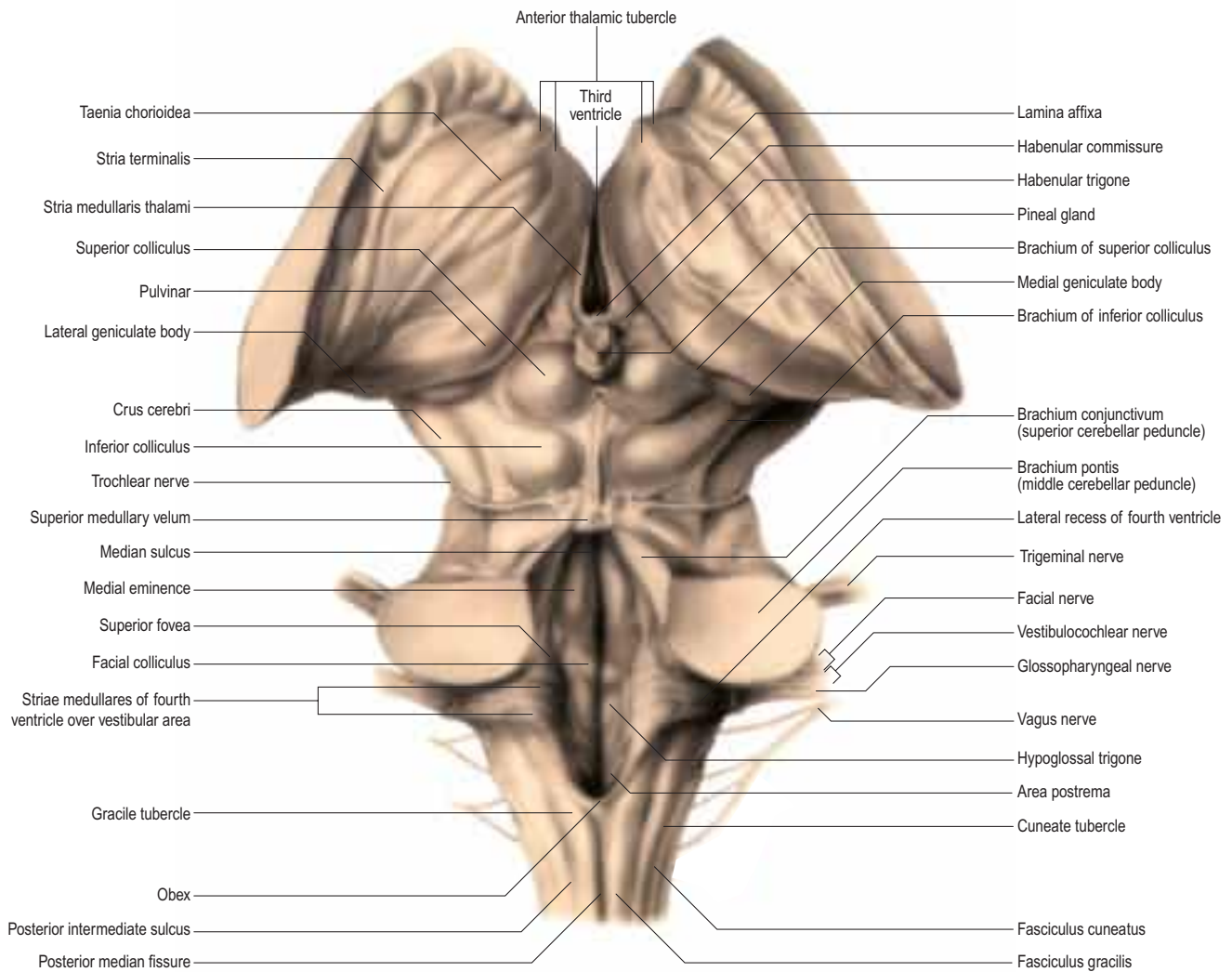


Fig. 21.5 The dorsal aspect of the brainstem. The floor of the fourth ventricle has been exposed by cutting the cerebellar peduncles and removing the cerebellum. (With permission from Neuroanatomy by FA Mettler (1948) 2nd edn; St Louis: The CV Mosby Company.)

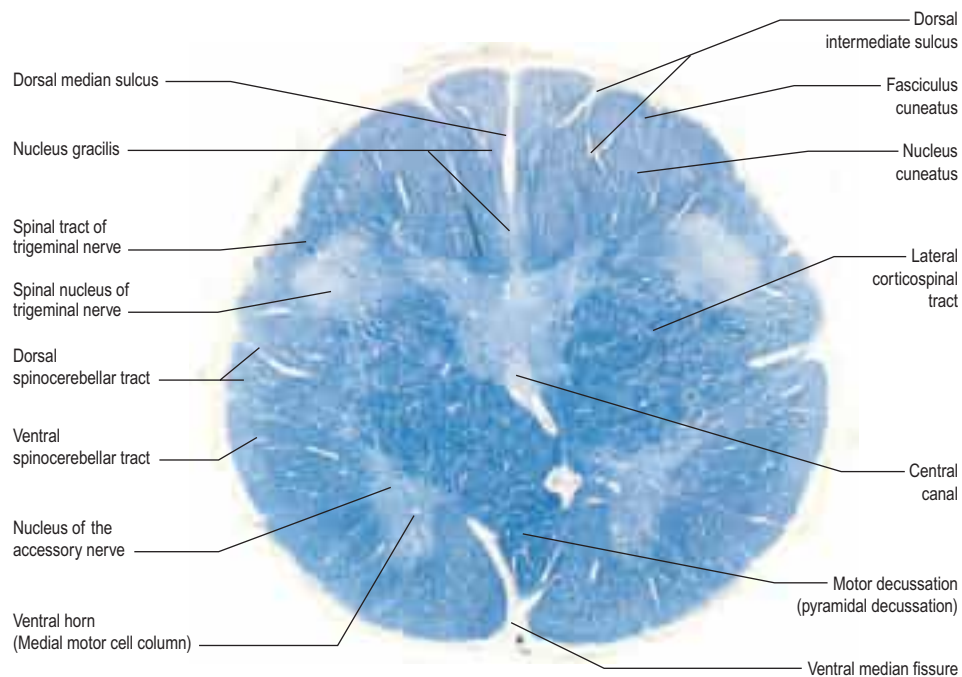


Fig. 21.6 A transverse section through the medulla oblongata at the level of the motor decussation. [Figures 21.6–21.8](#), [21.10–21.12](#) and [21.15–21.16](#) are all prepared using a stain for myelinated nerve fibres (blue).

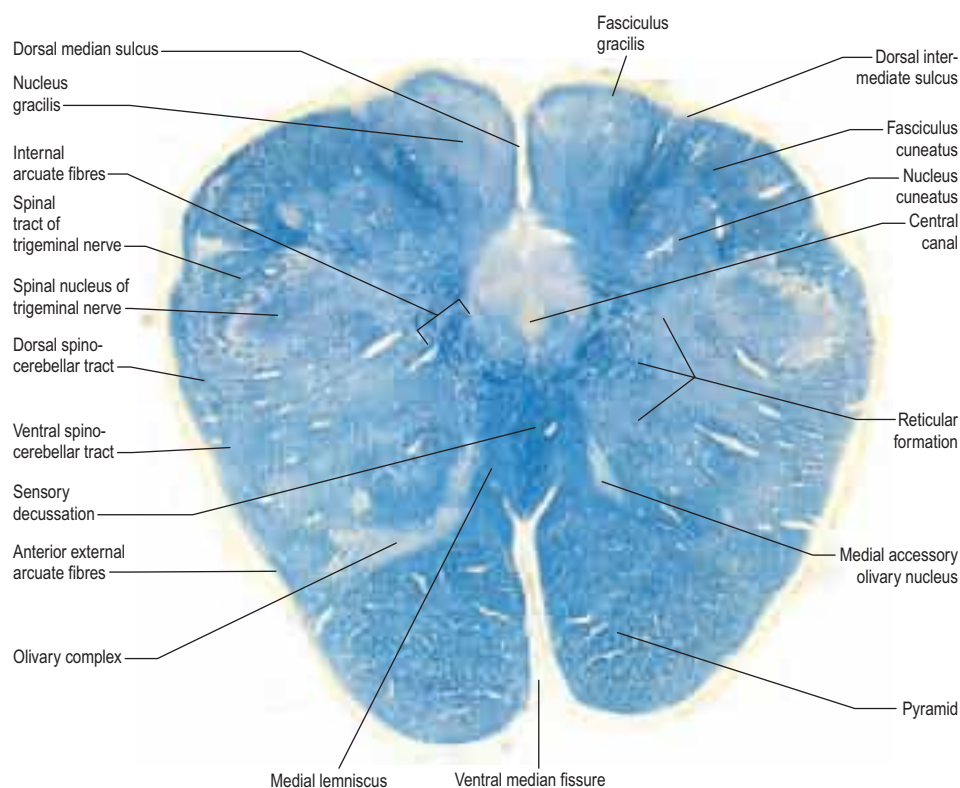


Fig. 21.7 A transverse section through the medulla oblongata at the level of the sensory decussation.

The dorsal grey column is also modified at this level as the nucleus gracilis appears in the ventral part of the fasciculus gracilis. The nucleus gracilis begins caudal to the nucleus cuneatus; the latter invades the fasciculus cuneatus from its ventral aspect in similar fashion.

The spinal nucleus and tract of the trigeminal nerve are visible ventrolateral to the dorsal columns and are continuous with the substantia gelatinosa and tract of Lissauer of the spinal cord.

Transverse section of the medulla at the level of the sensory decussation of the dorsal column–medial lemniscus system

The medullary white matter is rearranged above the level of the motor decussation (see Fig. 21.7). The pyramids form two large ventral bundles flanking the ventral median fissure on the ventral surface of the medulla; they contain corticospinal fibres of ipsilateral origin.

The nucleus gracilis is prominent on the dorsal medullary aspect, with diminishing numbers of fibres of the fasciculus gracilis located on its margins. The nucleus cuneatus is well developed. Both nuclei retain continuity with the central grey matter at this level but this is lost more rostrally. First-order afferent fibres contained within the fasciculi gracilis and cuneatus synapse on neurones in their respective nuclei (Millar and Basbaum 1975). Second-order axons emerge from the nuclei as internal arcuate fibres, at first curving ventrolaterally around the central grey matter and then ventromedially between the spinal tract of the trigeminal nerve and the central grey matter. The fibres decussate in the midline, as the sensory decussation, and thereafter form the medial lemniscus, which ascends to the thalamus (Haines 2012). The decussation of these internal arcuate fibres is located dorsal to the pyramids and ventral to the central grey matter, which is therefore more dorsally displaced than in the previous section.

The medial lemniscus ascends from the sensory decussation as a flattened tract near the median raphe. The pyramidal tract lies ventrally, and the medial longitudinal fasciculus and the tectospinal tract lie dorsally. Decussating internal arcuate fibres are rearranged as they cross, so that those derived from the nucleus gracilis lie ventral to those from the nucleus cuneatus. Above this level, the medial lemniscus is further rearranged so that gracile fibres migrate laterally, while cuneate fibres migrate medially. At pontine levels, the medial lemniscus is somatotopically organized with C1–S4 spinal segments represented sequentially from medial to lateral (Nieuwenhuys et al 2008).

The spinal nucleus of the trigeminal nerve is separated from the central grey matter by internal arcuate fibres, and from the lateral medullary surface by the spinal tract of the trigeminal nerve and by some dorsal spinocerebellar tract fibres. The latter progressively incline dor-

sally to enter the restiform body and then the inferior cerebellar peduncle at a higher level.

Transverse section of the medulla at the caudal end of the fourth ventricle

A transverse section at the caudal end of the fourth ventricle (Fig. 21.8) shows some new features together with most of those already described. The total area of grey matter is increased by the presence of the large inferior olivary nuclear complex and the nuclei of the vestibulocochlear, glossopharyngeal and vagus nerves.

A smooth, oval elevation, the olivary eminence, or olive, lies between the ventrolateral and dorsolateral sulci of the medulla. It is formed by the underlying inferior olivary complex and lies lateral to the pyramid, separated from it by the ventrolateral sulcus and emerging hypoglossal nerve fibres. The roots of the facial nerve emerge between its rostral end and the lower pontine border, in the cerebellopontine angle. The arcuate nuclei are curved, interrupted bands, located on the pyramids, and are said to be displaced pontine nuclei. Arcuatocerebellar fibres and those of the striae medullares are derived from them. They project mainly to the contralateral cerebellum through the inferior cerebellar peduncle (Fig. 21.9).

The inferior olivary complex (dorsal accessory, medial accessory, principal nuclei) is an irregularly crenated mass of grey matter with a medially directed hilum, through which numerous fibres enter and leave the nucleus (Nieuwenhuys et al 2008). It has prominent connections with the cerebellum and is described more fully in Chapter 22.

The central grey matter at this level constitutes the ventricular floor. It contains (sequentially from medial to lateral): the hypoglossal nucleus, dorsal motor nucleus of the vagus, nucleus solitarius, and the caudal ends of the inferior and medial vestibular nuclei.

The tractus and nucleus solitarius extend throughout the length of the medulla. The tract is composed of general visceral afferents from the vagus and glossopharyngeal nerves. The nucleus and its central connections with the reticular formation subserve the reflex control of cardiovascular, respiratory and cardiac functions (Ciriello 1983). The rostral portion of the nucleus solitarius (the gustatory nucleus) receives taste fibres from cranial nerves VII, IX and X. These same cranial nerves send general visceral sensation to the caudal portion of the nucleus solitarius (the cardiorespiratory nucleus).

The medial longitudinal fasciculus is a small compact tract near the midline, ventral to the hypoglossal nucleus. It contains fibre bundles including medial vestibulospinal tract fibres. At this medullary level, it is displaced dorsally by the pyramidal and lemniscal decussations. It ascends in the pons and midbrain, maintaining its relationship to the

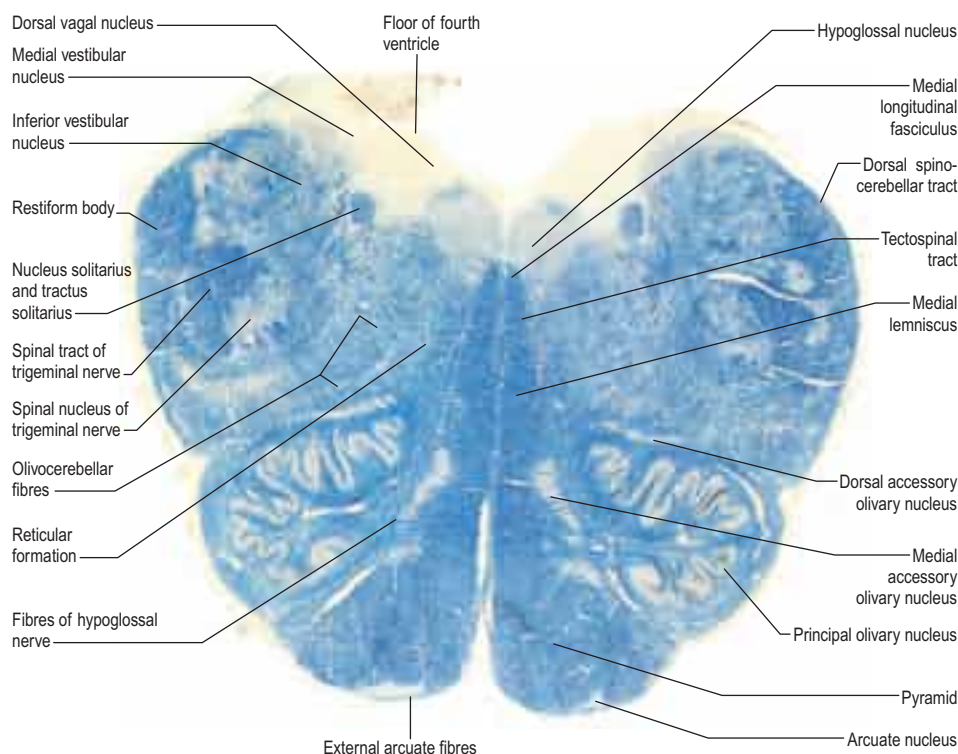


Fig. 21.8 A transverse section through the medulla oblongata at the caudal end of the fourth ventricle.

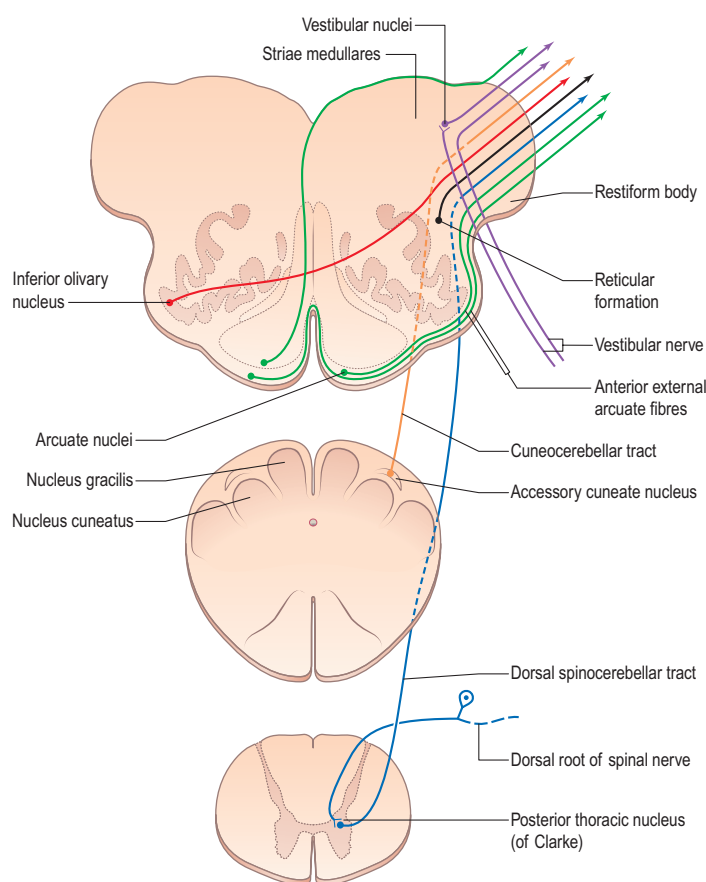


Fig. 21.9 Some of the afferent components of the restiform body. The efferent components have been omitted.

central grey matter and midline, and is therefore near the somatic efferent nuclear column. Fibres from a variety of sources course for variable distances in this tract.

The spinocerebellar, spinotectal, vestibulospinal and rubrospinal tracts and the anterolateral system (spinal lemniscal) all lie in the ventrolateral area of the medulla at this level (Nathan and Smith 1982). These tracts are limited dorsally by the spinal tract and nucleus of the trigeminal nerve, and ventrally by the pyramid.

Numerous islets of grey matter are scattered centrally in the ventrolateral medulla, an area intersected by nerve fibres that run in all

directions. This is the reticular formation, which exists throughout the medulla and extends into the pontine tegmentum and midbrain (Olszewski and Baxter 1954).

Corticospinal (pyramidal) tract

Each pyramid contains descending corticospinal fibres, derived from the ipsilateral cerebral cortex, which have traversed the internal capsule, midbrain and pons (see Fig. 20.16). Some 75–90% of the axons in the medullary pyramid cross the midline internal to the ventral median fissure as the motor (pyramidal) decussation. In the rostral part of the decussation, fibres cross ventromedially, whereas more caudally they pass dorsally, crossing ventral to the central grey matter. The decussation is orderly: fibres to cervical levels cross in its rostral part, whereas fibres to lumbosacral levels cross in its caudal portion. Fibres continue to pass dorsolaterally as they descend, to reach the contralateral lateral funiculus, where they form the crossed lateral corticospinal tract. A modest number of uncrossed corticospinal fibres descend in the ipsilateral ventral funiculus, as the ventral corticospinal tract, and a smaller uncrossed component joins the lateral corticospinal tracts (see Fig. 20.16). These small, uncrossed fascicles have no significant clinical relevance. The corticospinal tracts display somatotopy at almost all levels. In the pyramids the arrangement is like that at higher brainstem levels in that the lower extremity is represented laterally and the upper extremity is represented medially (Haines 2013, Nieuwenhuys et al 2008). Similar somatotopy is ascribed to the lateral corticospinal tracts within the spinal cord.

Dorsal column nuclei

The nuclei gracilis and cuneatus, the dorsal column nuclei, are part of the pathway, commonly called the dorsal column–medial lemniscus system, that is the major route for discriminative aspects of tactile and proprioceptive sensation (Haines 2015, Millar and Basbaum 1975, Nieuwenhuys et al 2008). The upper regions of both nuclei are reticular and contain small and large multipolar neurones with long dendrites. The lower regions contain clusters of large, round neurones with short and profusely branching dendrites. Upper and lower zones differ in their connections but both receive terminals from the dorsal spinal roots at all levels. Dorsal funicular fibres from neurones in the spinal grey matter are components of the postsynaptic dorsal column system and terminate only in the superior, reticular zone. Variable ordering and overlap of terminals, on the basis of spinal root levels, occur in both zones. The lower extremity is represented medially, the trunk ventrally and the digits dorsally. There is modal specificity, i.e. lower levels respond to low-threshold cutaneous stimuli, and upper reticular levels to inputs from fibres serving receptors in the skin, joints and muscles. The nucleus cuneatus is divided into several parts. Its middle zone contains a large pars rotunda, in which rostrocaudally elongated, medium-sized neurones are clustered between bundles of densely

myelinated fibres. The reticular poles of its rostral and caudal zones contain scattered, but evenly distributed, neurones of various sizes. The pars triangularis is smaller and laterally placed.

There is a somatotopic pattern of termination of cutaneous inputs from the upper limb on the cell clusters of the pars rotunda. Terminations are diffuse in the reticular poles.

The nucleus gracilis and nucleus cuneatus serve as relays between the spinal cord and higher levels. Primary spinal afferents synapse with multipolar neurones in the nuclei that form the major nuclear efferent projection. The nuclei also contain interneurones, many of which are inhibitory. Descending afferents from the somatosensory cortex reach the nuclei through the corticonuclear tracts, and appear to be restricted to the upper, reticular zones. These afferents both inhibit and enhance activity and are believed to be involved in sensory modulation. The reticular zones also receive connections from the reticular formation.

Neurones of the dorsal column nuclei receive terminals of long, uncrossed, primary afferent fibres of the fasciculi gracilis and cuneatus, which carry information concerning deformation of skin, movement of hairs, joint movement and vibration. Unit recording of the neurones in the dorsal column nuclei shows that their tactile receptive fields (i.e. the skin area in which a response can be elicited) vary in size, although they are especially small for the digits and the face and lips. Some fields have excitatory centres and inhibitory surrounds, which means that stimulation just outside its excitatory field inhibits the neurone. Neurones in the nuclei are spatially organized into a somatotopic map of the periphery (in accord with the similar localization in the dorsal columns). In general, specificity is high. Many cells receive input from one or a few specific receptor types, e.g. hair, type I and II slowly adapting receptors and Pacinian corpuscles, and some cells respond to Ia muscle spindle input. However, some neurones receive convergent input from tactile pressure and hair follicle receptors.

A variety of control mechanisms can modulate the transmission of impulses through the dorsal column–medial lemniscus pathway. Concomitant activity in adjacent dorsal column fibres may result in presynaptic inhibition by depolarization of the presynaptic terminals of one of them. Stimulation of the sensory–motor cortex also modulates the transmission of impulses by both pre- and postsynaptic inhibitory mechanisms, and sometimes by facilitation. These descending influences are mediated by the corticospinal tract. Modulation of transmission by inhibition also results from stimulation of the reticular formation, raphe nuclei and other sites.

The accessory cuneate nucleus, dorsolateral to the cuneate nucleus, is part of the spinocerebellar system of precerebellar relay nuclei (see Fig. 21.9). It contains large neurones like those in the posterior thoracic nucleus (Clarke's column) and receives the lateral fibres of the fasciculus cuneatus, which carry proprioceptive impulses from the upper limb that enter the cervical spinal cord rostral to the thoracic nucleus. The accessory cuneate neurones give rise to the posterior external arcuate

(cuneocerebellar) fibres that enter the cerebellum by the ipsilateral inferior cerebellar peduncle. A group of neurones, nucleus Z, identified in animals between the upper pole of the nucleus gracilis and the inferior vestibular nucleus, is said to be present in the human medulla. Its input is probably from the dorsal spinocerebellar tract, which carries proprioceptive information from the ipsilateral lower limb, and it projects through internal arcuate fibres to the contralateral medial lemniscus.

Sensory nucleus of the trigeminal nerve

The sensory nucleus of the trigeminal nerve is a large cell group that receives the primary afferents of the trigeminal nerve. It extends caudally into the cervical spinal cord and rostrally into the midbrain; its principal divisions are the spinal trigeminal, principal sensory and mesencephalic nuclei (see Figs 21.10–21.12) (Nieuwenhuys et al 2008; Olszewski 1950).

On entering the pons, the fibres of the sensory root of the trigeminal nerve run dorsomedially towards the principal sensory nucleus, which is situated at this level. Before reaching the nucleus, approximately 50% of the fibres divide into ascending and descending branches; the others ascend or descend without division. The descending fibres, of which 90% are less than 4 µm in diameter, form the spinal tract of the trigeminal nerve, which embraces the spinal nucleus of the trigeminal nerve and reaches the upper cervical spinal cord (see Figs 21.6–21.8; Fig. 21.10). There is a precise somatotopic organization within the tract (see Fig. 21.1). Fibres from the ophthalmic division of the trigeminal nerve lie ventrally, those from the mandibular division lie dorsally, and those from the maxillary division lie between. The tract is completed on its dorsal rim by fibres from the sensory roots of the facial, glossopharyngeal and vagus nerves. All of these fibres synapse in the pars caudalis of the spinal nucleus of the trigeminal nerve.

The detailed anatomy of the spinal tract of the trigeminal nerve excited early clinical interest because it was recognized that dissociated sensory loss could occur in the trigeminal area. For example, in Wallenberg's syndrome (lateral medullary syndrome), occlusion of the posterior inferior cerebellar artery (a branch of the vertebral artery) leads to loss of pain and thermal sense on the ipsilateral half of the face with retention of common sensation (Haines 2013). Neurosurgery in this region, as early as the 1890s, attempted to alleviate paroxysmal trigeminal neuralgia. The introduction of medullary tractotomy confirmed that dissociated thermoanalgesia of the face was associated with destruction of the tract.

There are conflicting opinions on the pattern of termination of the fibres in the spinal nucleus. It has long been held that fibres are organized rostrocaudally within the tract. According to this view, ophthalmic fibres are ventral and descend to the lower limit of the first cervical spinal segment, and maxillary fibres are central and do not extend below the medulla oblongata, whilst mandibular fibres are dorsal and do not

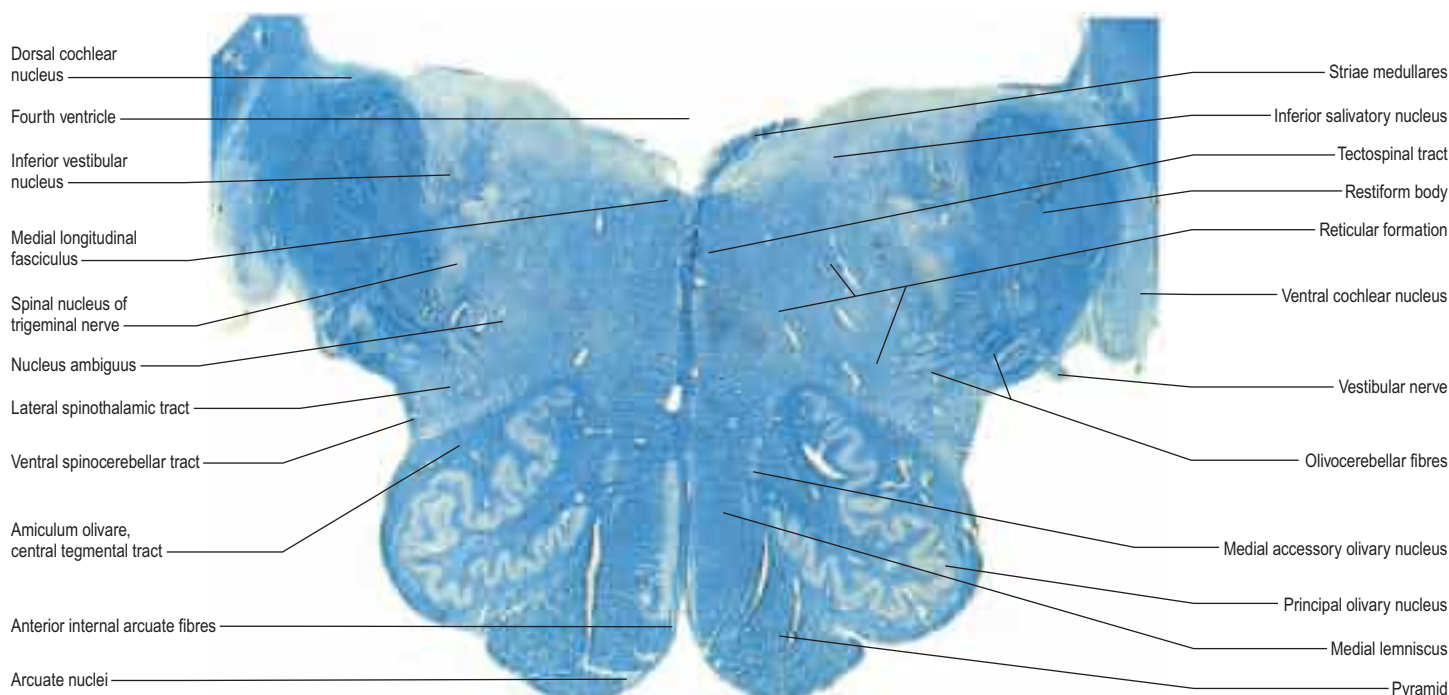


Fig. 21.10 A transverse section through the superior half of the medulla oblongata at the level of the inferior olivary nucleus.

extend much below the mid-medullary level. The results of section of the spinal tract in cases of severe trigeminal neuralgia support this distribution. It was found that a section 4 mm below the obex rendered the ophthalmic and maxillary areas analgesic, but tactile sensibility, apart from the abolition of 'tickles', was much less affected. To include the mandibular area it was necessary to section at the level of the obex.

More recently, it has been proposed that fibres are arranged dorso-ventrally within the spinal tract. There appear to be sound anatomico-physiological and clinical reasons for believing that all divisions terminate throughout the whole nucleus, although the ophthalmic division may not project fibres as far caudally as the maxillary and mandibular divisions. Fibres from the posterior face (adjacent to C2) terminate in the lower (caudal) part, whilst those from the upper lip, mouth and nasal tip terminate at a higher level. This can give rise to a segmental (cross-divisional) sensory loss in syringobulbia. Tractotomy of the spinal tract, if carried out at a lower level, can spare the perioral region, a finding that would accord with the 'onion-skin' pattern of loss of pain sensation. However, in clinical practice, the progression of anaesthesia on the face is commonly 'divisional' rather than strictly 'onion-skin' in distribution.

Fibres of the glossopharyngeal, vagus and facial nerves subserving common sensation (general somatic afferent) enter the dorsal region of the spinal tract of the trigeminal nerve and synapse with cells in the caudal part of the spinal nucleus. Consequently, operative section of the dorsal part of the spinal tract results in analgesia that extends to the mucosa of the tonsillar sinus, the posterior third of the tongue and adjoining parts of the pharyngeal wall (supplied by the glossopharyngeal nerve), and the cutaneous areas of the ear. Other afferents that reach the spinal nucleus are from the dorsal roots of the upper cervical nerves and from the sensory-motor cortex.

The spinal nucleus of the trigeminal nerve consists of three parts: the pars oralis (which adjoins the principal sensory nucleus); the pars interpolaris; and the pars caudalis (which is continuous with the dorsal horn of the spinal cord). The pars caudalis is different from the other parts because it has a structure analogous to that of the dorsal horn of the spinal cord, with a similar arrangement of cell laminae (subnuclei zonalis, gelatinosus and magnocellularis), and is involved in trigeminal pain perception. Cutaneous nociceptive afferents and small-diameter muscle afferents terminate in layers I, II, V and VI of the pars caudalis (see Fig. 21.1). Low-threshold mechanosensitive afferents of A β neurones terminate in layers III and IV of the pars caudalis and in the rostral (interpolaris, oralis, principal sensory) nuclei.

Many of the neurones in the pars caudalis that respond to cutaneous or tooth-pulp stimulation are also excited by stimulation of jaw or tongue muscles. This indicates that convergence of superficial and deep afferent inputs via wide-dynamic-range or nociceptive-specific neurones occurs in this nucleus. Similar convergence of superficial and deep inputs occurs in the rostral nuclei and may account for the poor localization of trigeminal pain, and for the spread of pain, which often makes diagnosis difficult.

There are distinct subtypes of cells in lamina II. Afferents from 'higher centres' arborize within it, as do axons from nociceptive and low-threshold afferents. Descending influences from these higher centres include fibres from the periaqueductal grey matter and from the nucleus raphe magnus and associated reticular formation.

The nucleus raphe magnus projects directly to the pars caudalis, probably via enkephalin, noradrenaline (norepinephrine) and 5-HT (5-hydroxytryptamine, serotonin)-containing terminals. These fibres directly, or indirectly through local interneurones, influence pain perception. Stimulation of periaqueductal grey matter or nucleus raphe magnus inhibits the jaw-opening reflex to nociception, and may induce primary afferent depolarization in tooth-pulp afferents and other nociceptive facial afferents. Neurones in the pars caudalis can be suppressed by stimuli applied outside their receptive field, particularly by noxious stimuli. The pars caudalis is an important site for relay of nociceptive input and functions as part of the pain 'gate control'. However, rostral nuclei may also have a nociceptive role. Tooth-pulp afferents via wide-dynamic-range and nociceptive-specific neurones may terminate in rostral nuclei, which all project to the subnucleus caudalis.

Most fibres arising in the trigeminal sensory nuclei cross the midline and ascend as trigeminothalamic fibres (trigeminal lemniscus). They end in the contralateral ventral posteromedial thalamic nucleus, from which third-order neurones project to the cortical postcentral gyrus (areas 3, 1, 2). However, some trigeminal nuclear efferents ascend to the ipsilateral ventral posteromedial nucleus.

Fibres from the pars caudalis, especially from laminae I, V and VI, also project to the rostral trigeminal nuclei, cerebellum, periaqueductal grey of the midbrain, parabrachial area of the pons, the brainstem reticular formation and the spinal cord. Fibres from lamina I project to the subnucleus medius of the medial thalamus.

Vagal nucleus

The vagal nucleus (the dorsal motor nucleus of the vagus) lies slightly dorsolateral to the hypoglossal nucleus, from which it is separated by the nucleus intercalatus. It extends caudally to the first cervical spinal segment and rostrally to the open part of the medulla under the vagal trigone in the floor of the fourth ventricle (see Fig. 21.8).

The vagal nucleus is a general visceral efferent nucleus and is the largest parasympathetic nucleus in the brainstem. Most (80%) of its neurones give rise to the preganglionic parasympathetic fibres of the vagus nerve. The remainder are interneurones or project centrally. The vagal nucleus innervates the non-striated (smooth, cardiac) muscle of the viscera of the thorax (heart, bronchi, lungs and oesophagus) and abdomen (stomach, liver, pancreas, spleen, small intestine and proximal part of the colon), and glandular epithelium. Neurones within the nucleus are heterogeneous and can be classified into nine subnuclei, which are regionally grouped into rostral, intermediate and caudal divisions. Topographic maps of visceral representation in animals suggest that the heart and lungs are represented in the caudal and lateral part of the nucleus, the stomach and pancreas in intermediate regions, and the remaining abdominal organs in the rostral and medial part of the nucleus.

There is a sparse sensory afferent supply, which arises in the nodose ganglion and projects directly to the nucleus and possibly beyond into the nucleus tractus solitarius.

Hypoglossal nucleus

The prominent hypoglossal nucleus lies near the midline in the dorsal medullary grey matter. It is approximately 2 cm long. Its rostral part lies beneath the hypoglossal trigone in the floor of the fourth ventricle (see Fig. 21.5) and its caudal part extends into the closed part of the medulla.

The hypoglossal nucleus consists of large motor neurones interspersed with myelinated fibres. It is organized into dorsal and ventral nuclear tiers, each divisible into medial and lateral subnuclei. There is a musculotopic organization of motor neurones within the nuclei that corresponds to the structural and functional divisions of tongue musculature. Thus, motor neurones innervating tongue retrusor muscles are located in dorsal/dorsolateral subnuclei, whereas motor neurones innervating the main tongue protrusor muscle are located in ventral/ventromedial regions of the nucleus. Although relatively little is known about the organization of motor neurones innervating the intrinsic muscles of the tongue, experimental evidence suggests that motor neurones of the medial division of the hypoglossal nucleus innervate tongue muscles that are orientated in planes transverse to the long axis of the tongue (transverse and vertical intrinsic and genioglossus), whereas motor neurones of the lateral division innervate tongue muscles that are orientated parallel to this axis (styloglossus, hyoglossus, superior and inferior longitudinal).

Hypoglossal fibres emerge ventrally from their nucleus, traverse the reticular formation lateral to the medial lemniscus, pass medial to the inferior olivary nuclei, and curve laterally to emerge as a linear series of 10–15 rootlets in the ventrolateral sulcus between the pyramid and olivary eminence (see Fig. 21.8).

The hypoglossal nucleus receives corticonuclear fibres from the precentral gyrus and adjacent areas of predominately the contralateral hemisphere. They synapse either on motor neurones of the nucleus directly or on interneurones. Evidence indicates that the most medial hypoglossal subnuclei may receive projections from both hemispheres. The nucleus may connect with the cerebellum via adjacent perihypoglossal nuclei, and perhaps also with the medullary reticular formation, the trigeminal sensory nuclei and the nucleus solitarius.

Inferior olivary nucleus

The olivary nuclear complex consists of a large principal olivary nucleus and smaller medial accessory and dorsal accessory olivary nuclei (see Figs 21.8, 21.10). They are also precerebellar nuclei, a group that includes the pontine, arcuate, vestibular, reticulocerebellar and spinocerebellar nuclei, all of which receive afferents from specific sources and project to the cerebellum. The inferior olivary nucleus contains small neurones, most of which form the olivocerebellar tract, which emerges primarily from the hilum to run medially and intersect the medial lemniscus (see Fig. 21.8). Its fibres cross the midline and sweep either dorsal to, or through, the opposite olivary nuclei. They intersect the lateral spinothalamic and rubrospinal tracts and the spinal nucleus of the trigeminal nerve, and enter the contralateral restiform body (and eventually the inferior cerebellar peduncle), where they constitute its major component. Fibres from the contralateral inferior olivary complex terminate on Purkinje cells in the cerebellum as climbing fibres; there is a one-to-one relationship between Purkinje cells and neurones in the complex (Nieuwenhuys et al 2008). Afferent connections to the inferior olivary nuclei are both ascending and descending. Ascending fibres,



Several smaller groups of cells lie near the hypoglossal nucleus (perihypoglossal nuclei), but none is known for certainty to be connected with the hypoglossal nerve or nucleus. They include the nucleus intercalatus, sublingual nucleus, nucleus prepositus hypoglossi and nucleus paramedianus dorsalis (reticularis). Gustatory and visceral connections are attributed to the nucleus intercalatus.

mainly crossed, arrive from all spinal levels in the spino-olivary tracts and via the dorsal columns. Descending ipsilateral fibres come from the cerebral cortex, cerebellar nuclei, thalamus, red nucleus and central grey of the midbrain. In part, the two latter projections make up the central tegmental tract.

Nucleus solitarius

The nucleus solitarius (solitary nucleus, nucleus of the solitary tract) lies lateral or ventrolateral to the vagal nucleus (see Fig. 21.8). A neuronal group ventrolateral to the nucleus solitarius has been termed the nucleus parasolitarius. The nucleus solitarius is intimately related to, and receives fibres from, the tractus solitarius, which carries afferent fibres from the facial, glossopharyngeal and vagus nerves (Ciriello 1983, Haines 2013, Hamilton and Norgren 1984). These fibres enter the tract in descending order and convey gustatory information from the lingual and palatal mucosa. They may also convey visceral impulses from the pharynx (glossopharyngeal and vagus) and from the oesophagus and abdominal alimentary canal (vagus). There is some overlap in this vertical representation.

The nucleus solitarius is thought to project to the sensory thalamus and thence to the cerebral cortex (Hamilton and Norgren 1984). It may also project to the upper levels of the spinal cord through a solitario-spinal tract. Secondary gustatory axons cross the midline. Many subsequently ascend the brainstem in association with the medial lemniscus and synapse on the most medial neurones of the ventral posteromedial thalamic nucleus (in a region sometimes termed the accessory arcuate nucleus). Axons from the ventral posteromedial nucleus radiate through the internal capsule to the antero-inferior area of the sensory-motor cortex and the insula. It is thought that other ascending paths end in a number of the hypothalamic nuclei, and so mediate the route by which gustatory information may reach the limbic system and allow appropriate autonomic reactions to be made.

Nucleus ambiguus

The nucleus ambiguus is a group of large motor neurones, situated deep in the medullary reticular formation (see Fig. 21.10). It extends rostrally as far as the upper end of the vagal nucleus, while caudally it is in line with, but is not continuous with, the nucleus of the accessory nerve. Fibres emerging from it pass dorsomedially, then curve laterally. Rostral fibres join the glossopharyngeal nerve. Caudal fibres join the vagus and are distributed to the pharyngeal constrictors, intrinsic laryngeal muscles and striated muscles of the palate and upper oesophagus.

The nucleus ambiguus receives corticonuclear fibres bilaterally with a contralateral preponderance and is connected to many brainstem centres. At its upper end, a small retrofacial nucleus intervenes between it and the facial nucleus. Although the nucleus ambiguus is generally regarded as a special visceral efferent nucleus, it is also a reputed source of general visceral efferent fibres to the vagus.

Swallowing and gag reflexes

Swallowing is initiated when food or liquid stimulates sensory nerves in the oropharynx and is usually regarded as programmed motor behaviour rather than a reflex. The patterning and timing of striated muscle contraction that occur in the pharynx, larynx and oesophagus during swallowing are generated in the brainstem in a network of neural circuits. The afferent limb is the glossopharyngeal nerve: information is relayed via the nucleus solitarius to the nucleus ambiguus, which contains the motor neurones innervating the muscles of the palate, pharynx and larynx. If stimulation of the oropharynx occurs other than during swallowing, a gag reflex may be initiated. There is a reflex contraction of the muscles of the pharynx, soft palate and fauces that, if extreme, may result in retching and vomiting.

Cough and sneeze reflexes

The cough reflex is normally initiated by irritation of tracheal or laryngeal mucosae: there is evidence of both mechanosensing and chemosensing cough receptors (McGarvey 2014). Coughing involves a sequence of coordinated events that produce the profound change in breathing pattern needed to expel an irritant from the lower airway. Rapid inspiration is followed by an expiratory effort against a closed glottis, the rapid generation of intrapulmonary pressure and the sudden opening of the glottis and contraction of intercostal and abdominal wall muscles, collectively producing a high-velocity flow of expired air that sweeps irritant material up towards the pharynx in a forceful exhalation (cough). Laryngeal branches of the vagus nerve carrying general visceral afferent information (with cell bodies in the inferior vagal ganglion) terminate in the nucleus of the solitary tract. Second-order neurones project to medullary respiratory centres (including a putative cough centre) and to the nucleus ambiguus, recruiting motor neurones innervating pharyngeal, laryngeal, diaphragmatic, intercostal and abdominal muscles.

Stimulation of nasal mucosa by physical or chemical irritants initiates a sneezing reflex. Afferent impulses travel via the ethmoidal and maxillary nerves to the spinal nucleus of the trigeminal nerve. Interneurones project to the nucleus ambiguus and a putative 'sneezing centre' in the rostral dorsolateral medulla (Seijo-Martínez et al 2006). Recruitment of a critical number of inspiratory and expiratory neurones initiates a sneeze, which involves eye closing and deep inspiration, followed by explosive exhalation as described above. If the oropharyngeal isthmus is closed by the action of palatoglossus, the air flow is diverted through the nasal cavity; otherwise the stream of expelled air flows through both oral and nasal cavities.

PONS

EXTERNAL FEATURES AND RELATIONS

The pons lies rostral to the medulla and caudal to the midbrain (see Fig. 21.4). On the ventral surface of the brainstem, the transition between medulla and pons is clearly demarcated by a transverse sulcus. Laterally, in a region known as the cerebellopontine angle, the facial, vestibulocochlear and glossopharyngeal nerves emerge. The ventral surface of the pons is separated from the clivus (basisphenoid and dorsum sellae) by the pontine cistern (cisterna pontis) (see Fig. 18.15). It is markedly convex transversely but less so vertically, and abuts the petrous part of the temporal bone laterally up to the internal acoustic meatus. The surface of the basilar pons has a shallow median sulcus, in which the basilar artery runs, bounded bilaterally by prominences formed partly by descending corticospinal fibres. Bundles of transverse fibres, bridging the midline and originating from nuclei in the basal pons (nuclei pontis, pontine nuclei), converge on each side into the large middle cerebellar peduncle and project to the cerebellum (Brodal and Bjaalie 1992, Haines 2013, Nieuwenhuys et al 2008). The trigeminal nerve emerges near the mid-pontine level; the basilar pons is inferior to this exit and the middle cerebellar peduncle superior and caudal to it. It has a small superomedial motor root (portio minor) and a large inferolateral sensory root (portio major).

The dorsal surface of the pons is hidden by the cerebellum, which covers the rostral half of the rhomboid fossa, into which the cerebral aqueduct empties. The roof of the fossa is formed by a thin sheet of tissue, the superior medullary velum, which is overlain by the lingula of the cerebellar vermis. The velum is attached on each side to the superior cerebellar peduncles and is enclosed by pia mater above and ependyma below (see Fig. 21.5). The trochlear nerves decussate in the rostral aspect of the velum.

INTERNAL ORGANIZATION

Transverse sections of the pons

Transverse sections (Figs 21.11–21.12) reveal that the pons consists of a tegmental part, which is a continuation of the medulla (excluding the pyramids), and a basilar part. The latter contains bundles of longitudinal descending fibres, some of which continue into the pyramids, while others end in the many pontine or medullary nuclei. It also contains numerous transverse fibres and pontine nuclei.

Basilar pons

The basilar pons is similar in structure at all levels. The longitudinal fibres of the corticopontine, corticonuclear and corticospinal tracts descend from the crus cerebri of the midbrain and enter the pons compactly. They rapidly disperse into fascicles, which are separated by the pontine nuclei and transverse pontine fibres. Corticospinal fibres run uninterrupted through the pons to the medullary pyramids, where they again converge into compact tracts. Corticonuclear fibres terminate in contralateral (and some ipsilateral) nuclei of cranial nerves and other nuclei in the pontine tegmentum. Clinical evidence supports the view that the neurones of the facial nucleus, and, in particular, those serving the muscles of the upper face, receive bilateral cortical control. The lower half of the face is controlled predominately by the contralateral cortex.

The pontine nuclei (see Fig. 21.11) include all the neurones (in humans, some 20 million) that are scattered throughout the basilar pons. They are probably all glutamatergic, and most project to the contralateral cerebellar cortex, with collaterals to the cerebellar nuclei. Efferent axons from the pontine nuclei constitute the transverse pontine (pontocerebellar) fibres, which decussate and continue as the major

The medial accessory olivary nucleus is a curved grey lamina, concave laterally, between the medial lemniscus and pyramid and the ventromedial aspect of the principal olivary nucleus. The dorsal accessory olivary nucleus is a similar lamina, dorsomedial to the principal olivary nucleus. Both accessory nuclei are connected with the cerebellum. These nuclei are phylogenetically older than the principal olivary nucleus, and are connected with the paleocerebellum. In all connections, cerebral, spinal and cerebellar, the olivary nuclei display specific topographical organization (Ch. 22).

Termination of special visceral afferents (taste) within the nucleus shows a viscerotopic pattern, predominantly in the rostral region (the gustatory nucleus). Experimental evidence suggests that fibres from the anterior two-thirds of the tongue and the roof of the oral cavity (which travel via the chorda tympani and greater petrosal branches of the facial nerve) terminate in the extreme rostral part of the solitary complex. Those from the circumvallate and foliate papillae of the posterior third of the tongue, tonsils, palate and pharynx (which travel via the lingual branch of the glossopharyngeal nerve) are distributed throughout the rostrocaudal extent of the nucleus, predominately rostral to the obex. Gustatory afferents from the larynx and epiglottis (which travel via the superior laryngeal branch of the vagus) have a more caudal and lateral distribution. The nucleus solitarius may also receive fibres from the spinal cord, cerebral cortex and cerebellum.

Medial and commissural subnuclei in the caudal part of the nucleus appear to be the primary site of termination for gastrointestinal afferents; this is the cardiorespiratory nucleus. Ventral and interstitial subnuclei probably receive tracheal, laryngeal and pulmonary afferents and play an important role in respiratory control and possibly rhythm generation. The carotid sinus and aortic body nerves terminate in the dorsal and dorsolateral region of the nucleus solitarius, which may be involved in cardiovascular regulation.

The nucleus ambiguus contains several cellular subgroups, and some topographical representation of the muscles innervated has been established. Individual laryngeal muscles are innervated by relatively discrete groups of cells in more caudal zones. Neurones that innervate the pharynx lie in the intermediate area, and neurones that innervate the oesophagus and soft palate are rostral.

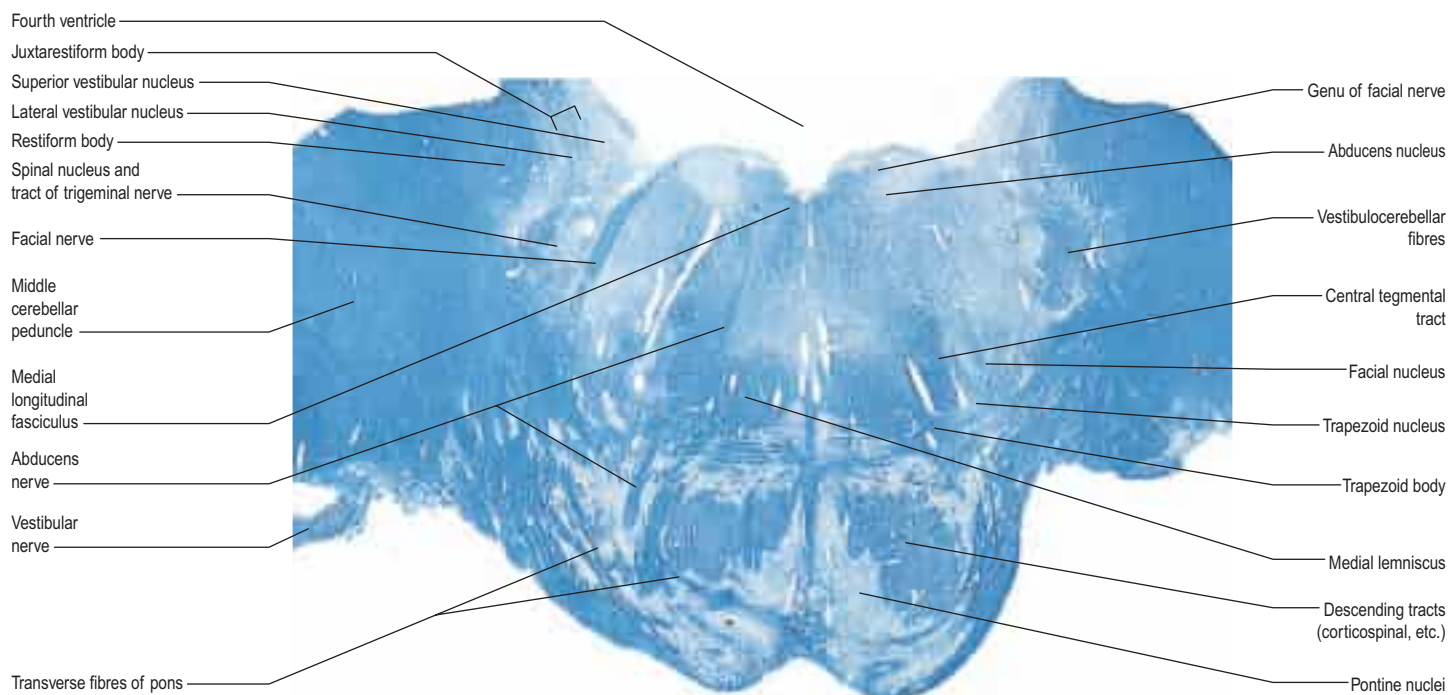


Fig. 21.11 A transverse section through the pons at the level of the facial colliculus.

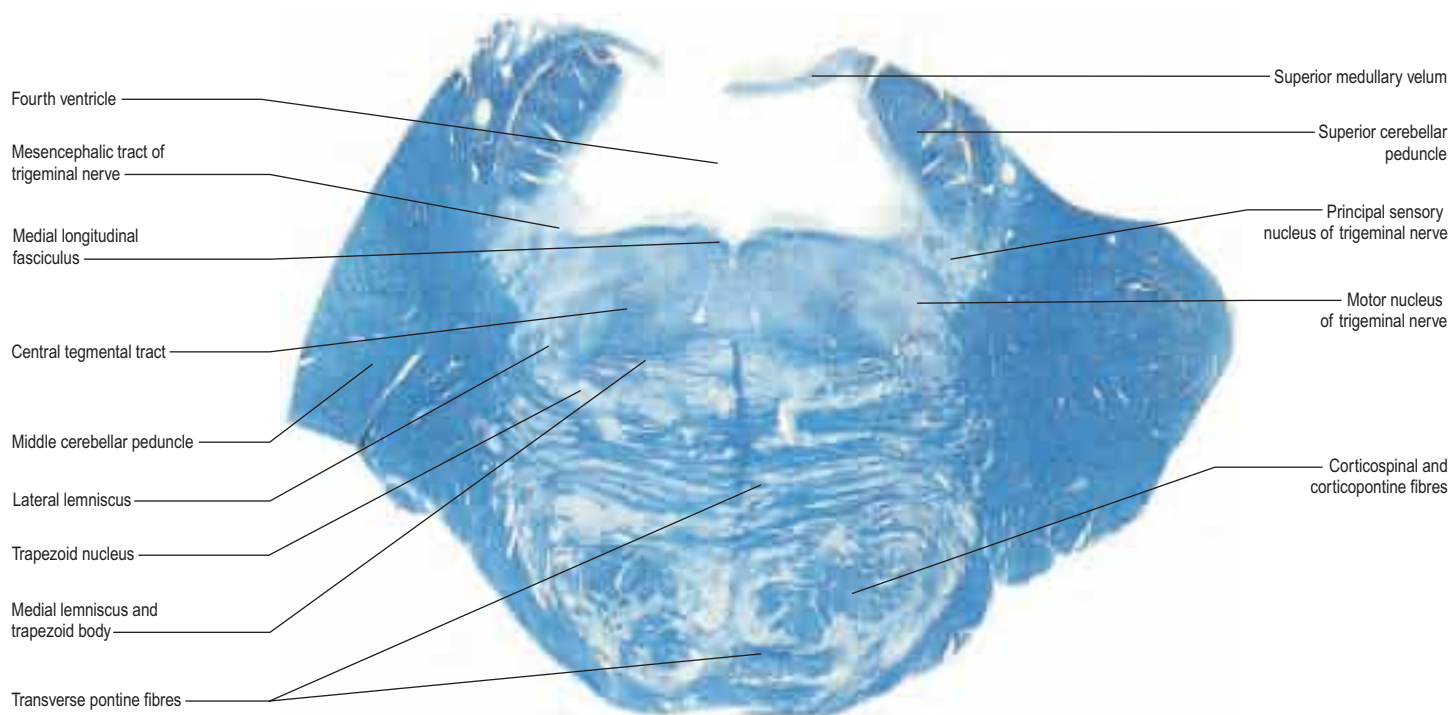


Fig. 21.12 A transverse section of the pons at the level of the principal sensory and motor nuclei of the trigeminal nerve.

constituent of the contralateral middle cerebellar peduncle. All pontocerebellar fibres end as mossy fibres in the cerebellar cortex, and a degree of somatotopy is maintained in these connections.

Corticopontine fibres, from the frontal, temporal, parietal and occipital cortices, end on neurones in the pontine nuclei. Frontopontine fibres end rostral to the level of the trigeminal nerve, and connect with pontine neurones that project to the contralateral cerebellum in the more rostral transverse pontine fibres. Corticopontine fibres arise mainly from neurones in layer V of the premotor, somatosensory, posterior parietal, extrastriate visual and cingulate neocortices. Projections from prefrontal, temporal and striate cortices are sparse. The terminal fields, although divergent, form topographically segmented patterns resembling overlapping columns, slabs or lamellae within the pons.

Subcortical projections to the pontine nuclei include those from the superior colliculus to the dorsolateral pons, and the medial mammillary nucleus to the rostromedial pons and pretectal nuclei. The lateral geniculate nucleus, dorsal column nuclei, trigeminal nuclei, hypothalamus and cerebellar nuclei also project to restricted neurones of the

pons. Functionally related subcortical and cerebrocortical afferents converge, e.g. those from the somatosensory cortex and dorsal column nuclei. There is also non-specific input from the reticular formation, raphe nuclei, locus coeruleus and periaqueductal grey matter.

Pontine tegmentum

The pontine tegmentum varies in cytoarchitecture at different levels (Haines 2015, Nieuwenhuys et al 2008, Olszewski and Baxter 1954). However, a consistent feature is that all cranial nerves associated with the pons have their nuclei (motor or sensory) within the tegmentum, although their axons may traverse the basilar pons. The pedunculopontine nucleus, as its name implies, lies in the caudal part of the tegmentum of the midbrain and extends into the rostral part of the dorsolateral tegmentum of the pons; it is described below.

A transverse section through the lower pontine tegmentum transects the facial colliculus in the floor of the fourth ventricle (see Fig. 21.11;

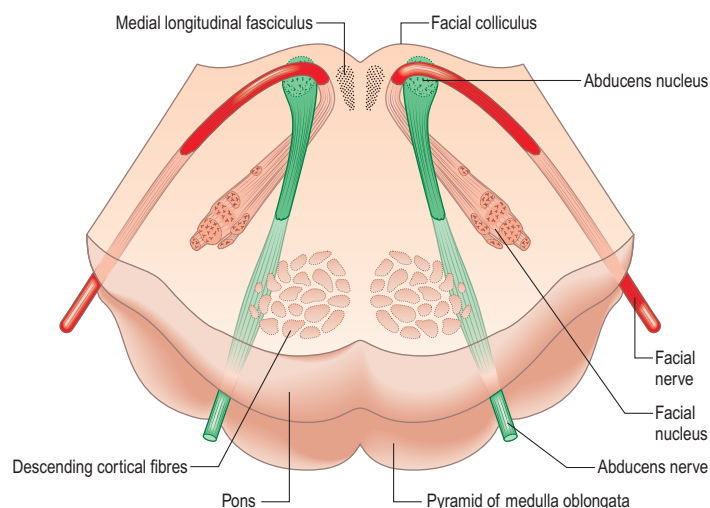


Fig. 21.13 A transverse section of the pons, viewed from the rostral aspect, showing the central course of fibres of the facial nerve.

Fig. 21.13). This colliculus contains the nucleus of the abducens nerve and the fibres of the internal genu of the facial nerve. More deeply placed are the facial nuclei, the nearby vestibular and cochlear nuclei, and other isolated neuronal groups. The vestibular nuclei are laterally placed in the rhomboid fossa of the fourth ventricle, subjacent to the vestibular area, which spans the rostral medulla and caudal pons (see Figs 21.3, 21.5).

The striae medullares of the fourth ventricle (see Fig. 21.10) are an aberrant cerebropontocerebellar connection, in which the arcuatocerebellar fibres are involved. Axons from arcuate nuclei spread round the medulla, above and below the olivary eminence, where they are superficially visible as the circumolivary fasciculus. All these fibres, collectively known as the external arcuate fibres, enter the restiform body (see Fig. 21.9). Some fibres from arcuate nuclei pass dorsally through the medulla near its midline (as arcuatocerebellar fibres), decussate near the floor of the fourth ventricle, then turn laterally under the ependyma and enter the cerebellum through the inferior peduncle via the restiform body.

In addition to the tracts already noted at lower levels, the lower pontine tegmentum contains the trapezoid body, lateral lemniscus and emerging fibres of the abducens and facial nerves. The medial lemniscus is ventral, its transverse outline now a flat oval. It extends laterally from the median raphe (see Figs 21.11, 21.12) and is laterally related to the anterolateral system and trigeminal lemniscus. The fibres of the latter originate from neurones of the contralateral spinal trigeminal nucleus, serving pain and thermal sensibility in facial, nasal and oral cavities and the conjunctiva. Here these fibres form a transverse band that is composed, in lateral order from the midline, of the medial lemniscus, trigeminal lemnisci, the anterolateral system and the lateral lemniscus.

The trapezoid body contains cochlear fibres, mainly from the ventral cochlear and trapezoid nuclei. They ascend transversely in the ventral tegmentum, pass either through or ventral to the medial lemniscal fibres, and decussate with the contralateral fibres in the median raphe. Below the emerging facial axons, the trapezoid fibres turn rostral to enter the lateral lemniscus. As the lateral lemniscus ascends, it lies near the dorsolateral surface of the brainstem. Rostrally, its fibres enter the inferior colliculus and medial geniculate body. The ascending auditory pathway is described in detail on [page 655](#).

The medial longitudinal fasciculus is paramedian, ventral to the fourth ventricle, near the midline and adjacent to motor nuclei of cranial nerves III, IV, VI and XII. It is the main intersegmental tract in the brainstem, particularly for interactions between nuclei of cranial nerves innervating the extraocular muscles and the vestibular system. In the lower pons it receives fibres from vestibular, and perhaps dorsal trapezoid, nuclei. Its greater part is formed by vestibular contributions.

A transverse section at an upper pontine tegmental level contains trigeminal elements (see Fig. 21.12), but otherwise shows little notable alteration from a section through a lower pontine tegmental level. Its dorsolateral parts are invaded by the superior cerebellar peduncles. The small nucleus of the lateral lemniscus, a relay in the auditory pathway, lies medial to its tract in the upper pons and receives some lemniscal terminals. Some of its efferent fibres enter the medial longitudinal fasciculus, while others return to the lemniscus.

The medial lemniscus (see Figs 21.11, 21.12) retains its position in the ventral pontine tegmentum, where it lies a little lateral to the median raphe, and is joined medially by fibres from the principal sensory nucleus of the trigeminal nerve. The trigeminal lemniscus, anterolateral system, and the lateral lemniscus and its nucleus all lie dorsolaterally.

Cochlear nuclei

Fibres of the cochlear division of the vestibulocochlear nerve partially encircle the restiform body laterally and end in the dorsal and ventral cochlear nuclei. The dorsal (posterior) cochlear nucleus forms a bulge, the auditory tubercle, on the posterior surface of the restiform body, and is continuous medially with the vestibular area in the rhomboid fossa. The ventral (anterior) cochlear nucleus is ventrolateral to the dorsal cochlear nucleus, and lies between the cochlear and vestibular fibres of the vestibulocochlear nerve.

Cochlear nerve fibres, which are derived from neuronal somata in the spiral ganglion, bifurcate on entering the brainstem and terminate in both dorsal and ventral cochlear nuclei (see Fig. 38.20). The ventral cochlear nucleus has a complex cytoarchitecture.

It contains many neuronal types with distinct dendritic field characteristics. Marked topographical order has been demonstrated in cochlear nerve terminals within the nucleus. Different parts of the spiral ganglion and differing stimulation frequencies are related to neurones that are serially arrayed anteroinferiorly in the ventral nucleus. All cochlear nerve fibres enter the nucleus. There are approximately 25,000 axons in the human cochlear nerve, and they project on to a much larger number of neurones in the cochlear nucleus. The number of cochlear fibres in the lateral lemniscus greatly exceeds that in the cochlear nerve. A minor fraction of the cochlear neurones receive terminals from the nerve, though each fibre may connect with several neurones. Terminals are limited to the anteroinferior region of the ventral nucleus, where the neurones are probably mostly local interneurones.

The dorsal (posterior) cochlear nucleus is generally continuous with the ventral nucleus, from which it is separated only by a thin stratum of nerve cell bodies and fibres. Giant cells predominate, and their dendritic fields are aligned with the incoming auditory fibres.

Though the cellular origins are not precisely known, axons of most neuronal types in the cochlear nuclei leave to end at pontine levels in the superior olivary, trapezoid and lateral lemniscal nuclei (see Fig. 21.12). They leave the cochlear nuclei by three routes. The largest group of axons lies ventrally and decussates as the trapezoid body, level with the pontomedullary junction (see Figs 21.11, 21.12). Most of these axons ascend slightly, decussate and relay in the contralateral nuclei. A few do not cross, and synapse in the ipsilateral superior olivary nuclei. From both nuclei, the next-order axons ascend in the corresponding lateral lemniscus. Occasional decussating fibres traverse the contralateral superior olive and enter the lateral lemniscus to relay in lemniscal nuclei.

Some axons from ventral cochlear neurones pass dorsally, superficial to descending trigeminal spinal fibres, cerebellar fibres in the restiform body and axons of the dorsal cochlear nucleus. This bundle of ventral cochlear fibres is smaller than that of the trapezoid decussation. It swerves ventromedially across the midline, ventral to the medial longitudinal fasciculus, as the intermediate acoustic striae. Its further path is uncertain, but it most likely ascends in the contralateral lateral lemniscus.

The most dorsally placed axons issue from the dorsal cochlear nucleus. They curve dorsomedially round the restiform body towards the midline as the dorsal acoustic striae, ventral to the striae medullares. They incline ventromedially and cross the midline to ascend in the contralateral lateral lemniscus, probably relaying in its nuclei.

The nuclei of the lateral lemniscus consist of small groups of neurones (counts of 18,000–24,000 have been recorded in humans) that lie among the fibres of the lateral lemniscus. Dorsal, ventral and intermediate groups probably receive afferent axons from both cochlear nuclei. Their efferents enter the midbrain along the lateral lemniscus and terminate in the inferior colliculi.

Some efferent axons travel in the cochlear nerves to the spiral organ; these comprise the olivocochlear tract (bundle of Rasmussen). Though few in number, they are involved in hearing, perhaps by modulating sensory transduction through reflexes via cochlear nuclei. The neurones of origin are located at the hilus and along the lateral border of the lateral superior olivary nucleus and lateral edge of the ventral trapezoid nucleus. Fibres from both sides proceed to both cochleae.

Vestibular nuclei

The vestibular complex consists of the medial (Schwalbe), lateral (Deiters' nucleus), superior (Bechterew) and inferior vestibular nuclei



The superior olivary complex is sited in the tegmentum of the caudal pons, lateral in the reticular formation at the level of the pontomedullary junction. This complex includes several named nuclei and nameless smaller groups. In humans, the lateral superior olivary nucleus is made up of some six small cellular clusters. The medial (accessory) superior olivary nucleus is large and compact. Cells in the immediate vicinity of these nuclei give rise to the olivocochlear tract. The trapezoid nucleus is medial and the retro-olivary group, also a potential source of cochlear efferents, is dorsal. Some internuclear connections have been described. The medial superior olivary nucleus receives impulses from both spiral organs, and may be involved in sound source localization. The superior olivary complexes and the trapezoid nuclei are relay stations in the ascending auditory projection. These intricate connections have not been definitively established in humans.

The medial nuclei of the trapezoid body are small in humans. They have a ventral component, which consists of large neurones scattered among the trapezoid fascicles, and a more compact dorsal nucleus, medial to the superior olivary complex. This nucleus lies at the level of the exiting abducens nerve roots, anterior to the central tegmental tract. It is likely that the human trapezoid nuclei function in the auditory relay pathway. Some trapezoid axons may enter the medial longitudinal fasciculus, and ascend to end in trigeminal, facial, oculomotor, trochlear and abducens nuclei, where they mediate reflexes involving tensor tympani, stapedius and oculogyric muscles, respectively.

(Haines 2013). The medial vestibular nucleus is the largest. It broadens, and then narrows as it ascends from the upper olivary level into the lower pons, where it is located lateral and superior to visceromotor nuclei of cranial nerves IX and X. It is crossed by the striae medullares, running nearer the floor, and is continuous with the nucleus intercalatus below. The inferior vestibular nucleus (which is the smallest) lies between the medial vestibular nucleus and restiform body, from the level of the upper end of the nucleus gracilis to the pontomedullary junction. It is traversed by descending fibres of the vestibular nerve, from the cerebellum, and the vestibulospinal tract. The lateral vestibular nucleus lies just above the inferior nucleus and is ventrolateral to the upper part of the medial nucleus. It ascends almost to the level of the abducens nucleus and its rostral end is continuous with the caudal end of the superior nucleus. The lateral vestibular nucleus is composed of large multipolar neurones, which are the main source of lateral vestibulospinal tract fibres. The superior vestibular nucleus is small and lies above the medial and lateral nuclei. It extends higher into the pons than other subdivisions, and occupies the upper part of the vestibular area.

Through its connections, the vestibular system influences movements of the eyes and head, and of the muscles of the trunk and limbs, in order to maintain equilibrium (Nieuwenhuys et al 2008). The vestibular nuclei receive fibres from the vestibular part of the vestibulocochlear nerve, cerebellum, brainstem reticular formation and spinal cord. They send axons to the cerebellum, medial longitudinal fasciculus, spinal cord and lateral lemniscus. Evidence suggests that the vestibular apparatus is spatially represented in the nuclei. Vestibular fibres of the vestibulocochlear nerve enter the medulla between the restiform body and the spinal tract of the trigeminal nerve, and approach the vestibular area, where they bifurcate into descending and ascending branches. The former descend medial to the restiform body and end in medial, lateral and inferior vestibular nuclei; the latter enter the superior and medial nuclei. A few vestibular fibres enter the cerebellum directly through the juxtarestiform body (the medial part of the inferior cerebellar peduncle), and end in the fastigial nucleus, flocculonodular lobe and uvula.

The vestibular nuclei project extensively to the cerebellum and also receive axons from the cerebellar cortex and the fastigial nuclei. Vestibulocerebellar fibres (primary and secondary) travel via the juxtarestiform body mainly to the flocculus and nodule. Cerebellovestibular fibres pass to the nuclei, also via the juxtarestiform body. They arise mainly in the flocculus and nodule (posterior lobe), but some fibres are derived from the anterior lobe and fastigial nucleus. The vestibular nuclear complex projects to the pontine reticular nuclei and to motor nuclei of the oculogyric muscles in the medial longitudinal fasciculus. The complex also projects to the spinal cord via medial and lateral vestibulospinal tracts, the former one part of the medial longitudinal fasciculus (see Fig. 20.18).

Information from the vestibular nuclei also reaches the cerebral cortex by way of the thalamus (probably via posterior parts of the ventroposterior complex and the medial pulvinar). The primary vestibular cortical area is located in the parietal lobe at the junction between the intraparietal and the postcentral sulci, which is adjacent to that portion of the postcentral gyrus where the head is represented. There may be an additional representation of the vestibular system in the superior temporal gyrus near the auditory cortex.

Abducens nucleus

The abducens nucleus is a group of approximately 6500 neurones, occupying a paramedian position in the central grey matter, in line with the trochlear, oculomotor and hypoglossal nuclei, with which it forms a somatic motor column (see Fig. 21.3). It lies laterally adjacent to the medial longitudinal fasciculus, which is the means by which vestibular, cochlear and other cranial nerve nuclei, especially the oculomotor, connect with the abducens. The abducens nucleus contains large motor neurones and small multipolar interneurones, which are intermixed, although the latter are most heavily concentrated in its lateral and ventral aspects. Axons of abducens interneurones cross the midline at the level of the nucleus and ascend in the medial longitudinal fasciculus to all three medial rectus subnuclei of the oculomotor nucleus.

Efferent abducens motor axons pass ventrally through the reticular formation, trapezoid body and medial lemniscus, and traverse the ventral pons to emerge at its inferior border (see Fig. 21.13); they innervate the lateral rectus muscle.

The abducens nucleus receives afferent connections from corticonuclear fibres (which are principally contralateral, some of the fibres being aberrant corticospinal fibres that descend from the midbrain to this level in the medial lemniscus); the medial longitudinal fasciculus (by which it is connected to oculomotor, trochlear and vestibular nuclei);

the tectobulbar tract (from the deep layers of the superior colliculus); the paramedian pontine reticular formation (which lies rostral and caudal to the nucleus); the nucleus prepositus hypoglossi; and the contralateral medullary reticular formation.

Facial nucleus

The facial (motor) nucleus lies in the caudal pontine reticular formation, posterior to the dorsal trapezoid nucleus and ventromedial to the spinal tract and nucleus of the trigeminal nerve (Haines 2013, Nieuwenhuys et al 2008). Groups of facial neurones form columns, which innervate individual muscles or which correspond to branches of the facial nerve. Neurones innervating muscles in the scalp and upper face are dorsal, and those supplying the lower facial musculature are ventral.

Efferent fibres of the large motor neurones of the facial nucleus form the motor root of the facial nerve. The motor nucleus is part of the pharyngeal efferent column, but it lies much more deeply in the pons than might be expected and its axons have an unusual course (see Fig. 21.13). At first they incline dorsomedially towards the fourth ventricle, caudal to the abducens nucleus, and ascend medial to it, near the medial longitudinal fasciculus. They then curve round the rostral pole of the abducens nucleus and descend ventrolaterally through the reticular formation. Finally, they pass between their own nucleus medially and the spinal nucleus of the trigeminal nerve laterally. They emerge between the olive and restiform body at the cerebellopontine angle.

The facial nucleus receives corticonuclear fibres for volitional control. Neurones that innervate muscles in the scalp and upper face receive bilateral corticonuclear fibres, while those supplying lower facial musculature receive a predominately contralateral innervation. Clinically, upper and lower motor neurone lesions of the facial nerve can be differentiated because the former results in paralysis that is confined to the contralateral lower face (supranuclear facial palsy), whilst the latter results in a complete ipsilateral paralysis (Bell's palsy).

The facial nucleus also receives ipsilateral afferents from the nucleus solitarius and from the sensory nucleus of the trigeminal nerve, establishing important reflex connections.

Some efferent fibres of the facial nerve originate from visceromotor neurones in the superior salivatory nucleus, which is in the reticular formation dorsolateral to the facial nucleus. These preganglionic parasympathetic neurones belong to the general visceral efferent column. They send fibres into the sensory root of the facial nerve, a part of the intermediate nerve. These fibres travel via the chorda tympani to the submandibular ganglion and via the greater petrosal nerve and the nerve of the pterygoid canal to reach the pterygopalatine ganglion.

Corneal and blink reflexes

Touching the cornea of one eye elicits reflex closure of both eyes via the corneal reflex. The afferent limb of the pathway involves mainly nasociliary branches of the ophthalmic division of the trigeminal nerve, with cell bodies in the trigeminal ganglion and central processes synapsing on second-order neurones in the ipsilateral spinal tract and chief sensory nuclei of the trigeminal nerve. These neurones drive a polysynaptic chain of interneurones in the lateral reticular formation in the pons and medulla, which, in turn, activate neurones innervating the palpebral components of orbicularis oculi in both ipsilateral and contralateral facial motor nuclei; the stimulated muscles contract to produce a bilateral blink. The sweep of the eyelids carries lacrimal secretions across the eye, helping to remove any irritating particles.

Shining a bright light into one eye elicits a similar reflex closure of both eyes via the blink reflex; the afferent limb of this reflex involves stimulation of the retina and optic pathway.

Trigeminal sensory nucleus

On entering the pons, the fibres of the sensory root of the trigeminal nerve run dorsomedially towards the principal sensory nucleus (see Fig. 21.12). About 50% of the fibres divide into ascending and descending branches; the others ascend or descend without division. The descending fibres form the spinal tract of the trigeminal nerve, which terminates in the medially adjacent spinal nucleus of the trigeminal nerve.

Some ascending trigeminal fibres, many of them heavily myelinated, synapse around the small neurones in the principal sensory nucleus (see Fig. 21.12), which lies lateral to the motor nucleus and medial to the middle cerebellar peduncle, and is continuous inferiorly with the spinal nucleus of the trigeminal nerve. The principal nucleus is considered to be mainly concerned with tactile stimuli.

Other ascending fibres enter the mesencephalic nucleus (Fig. 21.14). This column of cells contains the only primary sensory unipolar neurones with somata in the CNS rather than in a sensory ganglion. The

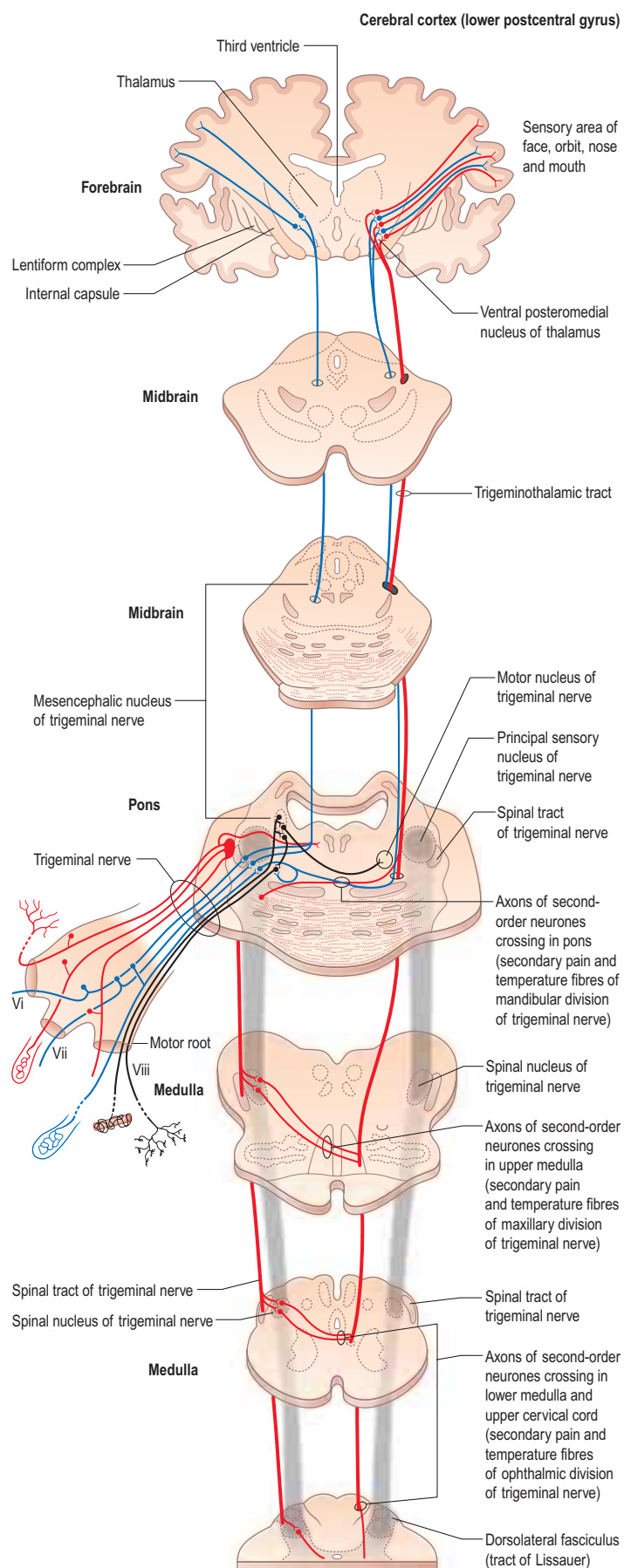


Fig. 21.14 The trigeminal nerve and its central connections.

peripheral processes of these neurones may convey proprioceptive impulses from the masticatory muscles, and possibly also from the teeth and the facial and oculogyric muscles. The nucleus is the relay for the 'jaw jerk', the only significant supraspinal monosynaptic reflex, elicited by lightly tapping the relaxed open jaw in a downward direction. Nerve fibres that ascend to the mesencephalic nucleus may give collaterals to the motor nucleus of the trigeminal nerve and to the cerebellum.

Jaw jerk reflex

Rapid stretching of the muscles that close the jaw (masseter, temporalis, medial pterygoid) activates muscle spindle afferents, which travel via the mandibular division of the trigeminal nerve to the brain stem. The cell bodies of these primary afferent neurones are located in the mesencephalic trigeminal nucleus. Collaterals project monosynaptically to the motor nucleus of the trigeminal nerve in the pons. Motor axons arising in the nucleus travel via the mandibular nerve to innervate the muscles that act on the temporomandibular joint and close the jaw (Chapter 32).

Most fibres that arise in the trigeminal sensory nuclei cross the midline and ascend in the trigeminal lemniscus. They end in the contralateral ventral posteromedial thalamic nucleus, from which third-order neurones project to the postcentral gyrus (areas 3, 1, 2). Some trigeminal nuclear efferents ascend to the ventral posteromedial nucleus of the thalamus.

Trigeminal motor nucleus

The motor nucleus of the trigeminal nerve is ovoid in outline and lies in the upper pontine tegmentum, under the lateral part of the floor of the fourth ventricle (see Fig. 21.3). It lies medial to the principal sensory nucleus, and is separated from it by fibres of the trigeminal nerve. It forms the rostral part of the pharyngeal (special visceral) efferent column.

The motor nucleus contains characteristic large multipolar neurones interspersed with smaller multipolar cells. The neurones are organized into a number of relatively discrete subnuclei, the axons from which innervate individual muscles. It receives fibres from both corticonuclear tracts. These fibres leave the tracts at the nuclear level or higher in the pons (aberrant corticospinal fibres), and may end on motor neurones or interneurones. The motor nucleus receives afferents from the sensory nuclei of the trigeminal nerve, possibly including some from the mesencephalic nucleus, which form monosynaptic reflex arcs for proprioceptive control of the masticatory muscles. It also receives afferents from the reticular formation, red nucleus and tectum, the medial longitudinal fasciculus and possibly from the locus coeruleus. Collectively, these represent pathways by which salivary secretion and mastication may be coordinated.

Tensor tympani and stapedius reflex

Loud sound elicits reflex contraction of tensor tympani and stapedius, which attenuates movement of the tympanic membrane and middle ear ossicles. Afferent impulses travel in the cochlear nerve to the cochlear nuclei in the brainstem. Efferent fibres to tensor tympani arise in the motor nucleus of the trigeminal nerve and travel in the mandibular division of the nerve. Efferent fibres to stapedius originate in the facial nucleus and travel in the facial nerve.

Salivatory nuclei

The salivatory nuclei are near the upper pole of the vagal nucleus, just above the pontomedullary junction and near the inferior pole of the facial nucleus. They are divided into superior and inferior salivatory nuclei, which send preganglionic parasympathetic fibres into the facial and glossopharyngeal nerves, respectively, for the control of the salivary and lacrimal glands.

MIDBRAIN

EXTERNAL FEATURES AND RELATIONS

The midbrain traverses the tentorial notch and connects the pons and cerebellum with the diencephalon. It is the shortest brainstem segment, not more than 2 cm in length, and much of it lies in the posterior cranial fossa (Haines 2013, Nieuwenhuys et al 2008). Lateral to it are the parahippocampal gyri, which hide its sides when the inferior surface of the brain is examined. For descriptive purposes, the midbrain may be divided into a tectum containing the pretectal area and corpora quadrigemina, and a basis pedunculi consisting of the substantia nigra and the crus cerebri (see Fig. 21.2). The former contains pigmented cells and the latter clinically important descending tracts. The midbrain tegmentum is the central midbrain area between the tectum and the substantia nigra; the cerebral aqueduct is located at the interface of tectum and tegmentum within the periaqueductal grey (Haines 2013). The cerebral peduncle may be regarded as the midbrain located below the tectum (the tegmentum plus basis pedunculi).

The crura cerebri are superficially corrugated and emerge from the cerebral hemispheres. They converge as they descend, forming the

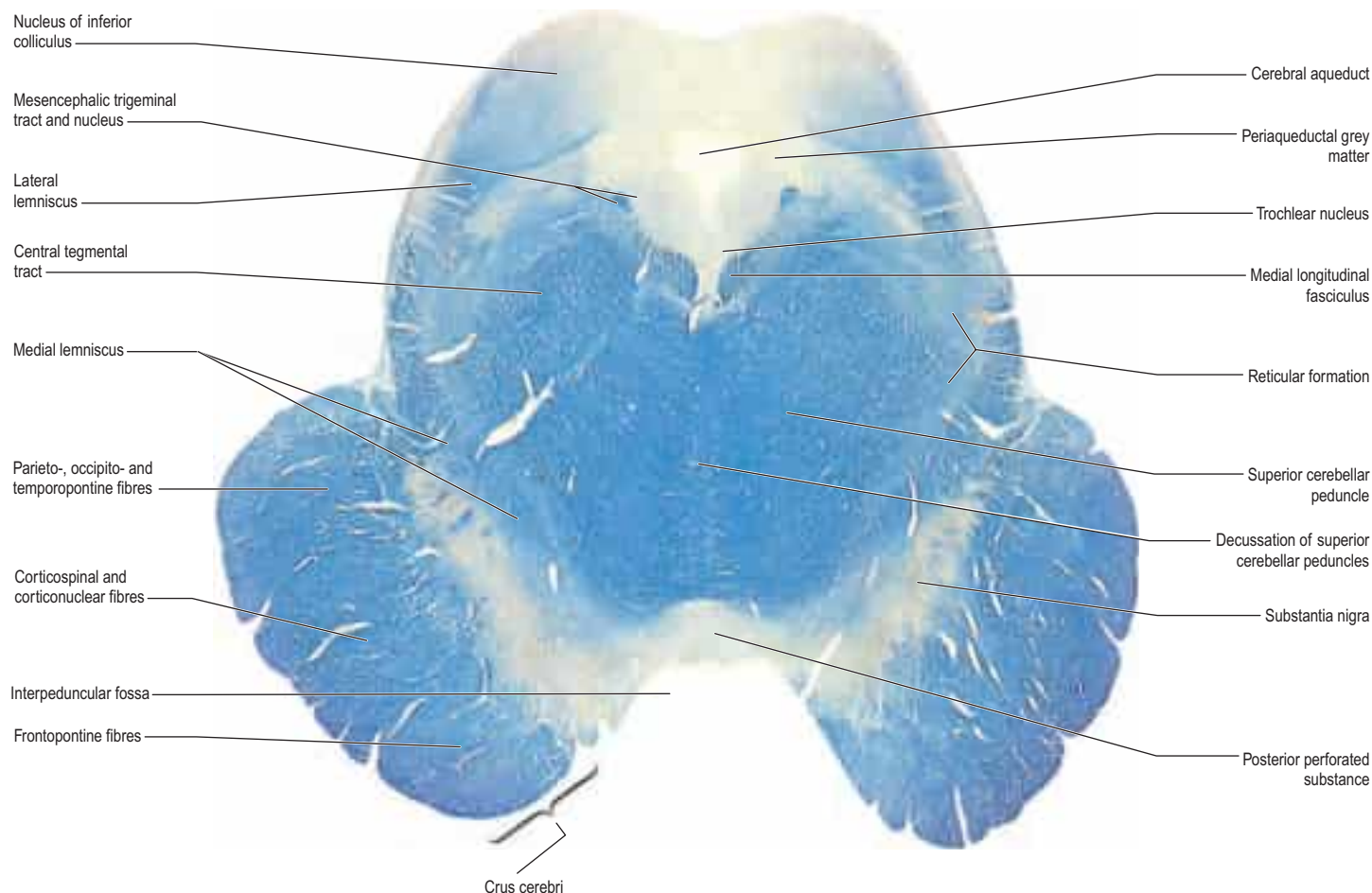


Fig. 21.15 A transverse section of the midbrain at the level of the inferior colliculi.

caudolateral boundaries of the interpeduncular fossa, and eventually enter the basilar pons (Figs 21.15–21.16). At the level of the tentorial incisura, the basilar artery divides in the interpeduncular fossa into the right and left P_1 segments of the posterior cerebral arteries. The superior cerebellar arteries branch from the basilar artery immediately before this bifurcation. The posterior cerebral and superior cerebellar arteries both run laterally around the ventral surface of the crus, the former passing above the tentorium cerebelli, the latter passing below. The oculomotor and trochlear nerves lie between these two arteries. The roots of the oculomotor nerve emerge from the medial aspect of the crus via the oculomotor sulcus. The posterior communicating artery joins the posterior cerebral artery on the medial surface of the peduncle in the interpeduncular fossa. The numerous small openings in the interpeduncular fossa constitute the posterior perforated substance, which is pierced by central branches of the posterior cerebral arteries. The optic tract is located at the point where the crus emerges from the cerebral hemispheres.

The superior and inferior colliculi are two paired eminences on the dorsal surface of the midbrain (see Fig. 21.5). They lie rostral to the superior medullary velum, inferior to the pineal gland and caudal to the posterior commissure. Below the splenium of the corpus callosum, they are inferior and medial to the pulvinar of the thalamus. The superior and inferior colliculi are separated by a cruciform sulcus. The upper limit of the sulcus expands into a depression for the pineal gland, and a median frenulum veli is prolonged from its caudal end down over the superior medullary velum. The trochlear nerve emerges lateral to the frenulum and caudal to the inferior colliculus, passes ventrally over the lateral aspects of the cerebral peduncles and traverses the interpeduncular cistern to the petrosal end of the cavernous sinus. The superior colliculi, larger than the inferior, are centres for visual reflex responses. The inferior colliculi are a part of the ascending auditory pathway.

A brachium passes ventrolaterally from the lateral aspect of each colliculus (see Fig. 21.5). The brachium of the superior colliculus (superior quadrigeminal brachium) passes below the pulvinar, partly overlapping the medial geniculate body, and conveys fibres from the retina to the superior colliculus. The brachium of the inferior colliculus (inferior quadrigeminal brachium) ascends ventrally and conveys fibres from the lateral lemniscus and inferior colliculus to the medial geniculate body.

INTERNAL ORGANIZATION

Transverse sections of the midbrain

In transverse section, the cerebral peduncles are seen to be composed of dorsal and ventral regions separated by the substantia nigra (see Figs 21.15–21.16). On each side, the dorsal region is the tegmentum and the ventral part is the crus cerebri. The tegmenti are continuous across the midline but the crura are separated.

Crus cerebri

The crus cerebri is approximately semilunar in section, being convex externally and concave internally. It contains corticonuclear, corticospinal and corticopontine fibres. Corticonuclear fibres terminate at various brainstem levels, many in association with the cranial nerve nuclei. Corticospinal fibres pass uninterrupted through the brainstem to enter the medullary pyramid and, thereafter, form the corticospinal tracts of the spinal cord (see Fig. 20.17). Corticonuclear and corticospinal fibres occupy about the middle two-thirds of the crus cerebri. Corticopontine fibres arise in the cerebral cortex and form two groups, both of which end in the pontine nuclei. Frontopontine fibres from the frontal lobe, principally areas 6 and 4, traverse the internal capsule and then occupy the medial fourth of the ipsilateral crus cerebri. Temporopontine fibres, which originate largely from the posterior region of the temporal lobe, traverse the internal capsule and then run in the lateral part of the ipsilateral crus. Parietopontine and occipitopontine fibres are also described in the crus, lying medial to the temporopontine fibres; collectively, these fibres occupy about the lateral third of the crus. There are few fibres from the primary sensory cortex in corticopontine projections.

Substantia nigra

The substantia nigra is a large nucleus that extends through the entire length of the midbrain, located between the crus cerebri ventrolaterally and ascending lemniscal fibres dorsomedially (see Figs 21.15–21.16). Its medial part is traversed by oculomotor axons passing to their point

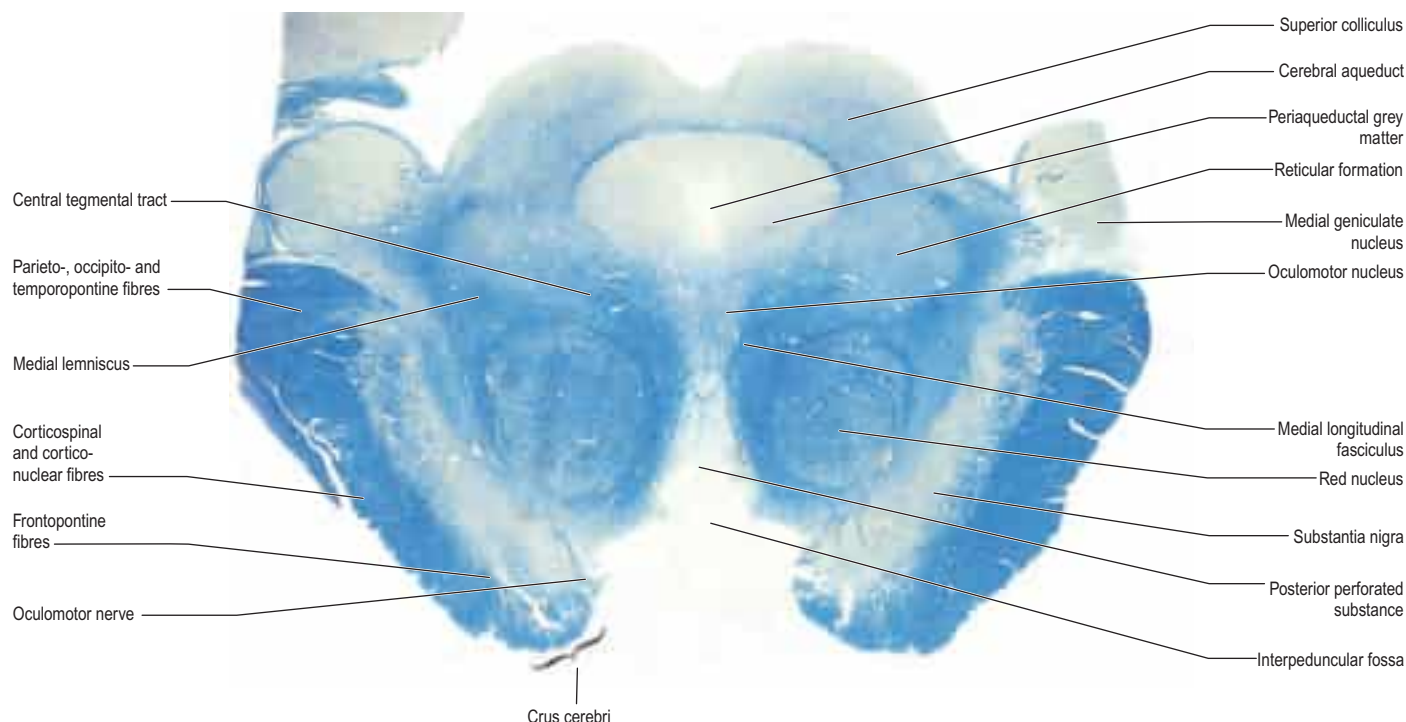


Fig. 21.16 A transverse section of the midbrain at the level of the superior colliculi.

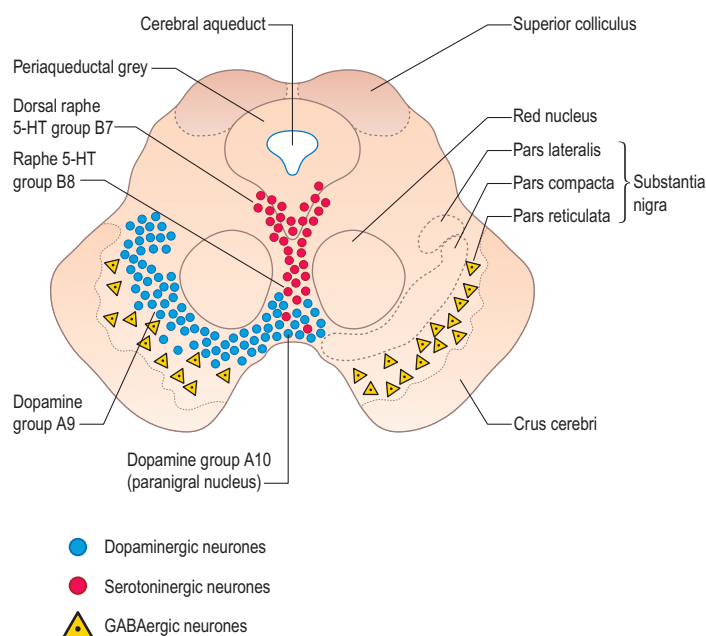


Fig. 21.17 A transverse section through the midbrain to show the arrangement of dopaminergic cell groups A9 and A10 in the substantia nigra (left) and serotonergic cell groups B7 and B8 in the raphe.

of exit in the interpeduncular fossa. The substantia nigra is divided into a larger part consisting of a dorsal pars compacta and a ventral pars reticularis and a smaller pars lateralis.

The pars compacta consists of darkly pigmented neurones, which contain neuromelanin granules. Their arrangement is irregular and they partially penetrate the subjacent pars reticularis. Pigmentation increases with age, is most abundant in primates, maximal in humans, and present even in albinos. The pigmented neurones synthesize dopamine as their neurotransmitter and correspond to the dopaminergic cell group A9 (Fig. 21.17). They project predominately to the caudate nucleus and putamen in a topographically organized fashion, but also project to the globus pallidus and subthalamic nucleus. Together with the retrorubral nucleus (A8), they comprise most of the dopaminergic neurone population of the midbrain and are the source of the mesostriatal dopamine system that projects to the striatum. The pars compacta of each side is continuous with its contralateral counterpart through the ventral tegmental dopamine group A10, which is sometimes known as the paranigral nucleus. This is the source of the

mesolimbic dopamine system, which supplies the ventral striatum, neighbouring parts of the dorsal striatum, and the prefrontal and anterior cingulate cortices. The dopaminergic neurones of the pars compacta (A9) and adjacent ventral tegmental area (A10) also contain cholecystokinin (CCK) or somatostatin.

The pars reticularis extends rostrally as far as the subthalamic region. It consists of clusters of large multipolar neurones, many of which intermingle with fibres of the crus cerebri. It is considered to be a homologue of the internal (medial) segment of the globus pallidus, which it resembles in terms of cell type and connectivity. It contains neurones that are very similar to those of the globus pallidus; their disc-like dendritic trees, like those of the pallidum, are orientated at right angles to afferents from the striatum, probably making *en passant* contacts. Most pars reticularis neurones are GABAergic and also contain high levels of iron. They project rostrally to the ventral anterior and dorsomedial thalamic nuclei and the deep (polysensory) layers of the superior colliculus, and caudally to the pedunculopontine nucleus and brainstem reticular formation, from where impulses are relayed to spinal ventral horn neurones. The connection from the striatum to the superior colliculus, via the pars reticularis, is thought to function in the control of gaze in a manner analogous to the pathway that initiates general body movement via the pallidum, thalamus and supplementary motor cortex. The uncontrolled or fixed-gaze disturbances of advanced Parkinson's disease, progressive supranuclear palsy and Huntington's disease tend to support this.

There are reciprocal connections between the substantia nigra and the basal ganglia, and these are discussed in Chapter 24. Briefly, topographically organized striatonigral axons, some of which may be collaterals of striatopallidal axons, originate from the caudate nucleus and putamen and project to the pars reticularis. They utilize GABA and substance P or enkephalin, and are distributed differentially in the pars reticularis in that enkephalinergic axons terminate in the medial part, whereas substance P axons terminate throughout. The head of the caudate nucleus projects to the rostral third of the substantia nigra, while the putamen projects to all parts. The fibres end in axodendritic synapses. A small number of GABAergic pallidonigral fibres from the external (lateral) segment of the globus pallidus end mostly in the pars reticularis. The subthalamic nucleus sends an important glutamatergic projection to the pars reticularis and to the globus pallidus. Some corticonigral fibres appear to exist, passing from precentral and postcentral gyri to neurones in the pars reticularis: this is predominantly a glutamatergic projection.

The pars lateralis projects to the ipsilateral superior colliculus, and may be involved in the control of saccadic eye movements.

Two other dopaminergic cell groups are found in the ventral tegmentum: cell group A10 in the rostromedial region, which corresponds to the ventral tegmental area (of Tsai), and cell group A8 in the dorsolateral reticular area, which forms the nucleus parabrachialis pigmentosus

(Nieuwenhuys 1985). The whole ventral tegmental system of dopaminergic neurones acts as an integrative centre for adaptive behaviour. It projects via a number of pathways, mainly through ascending fibres in the ipsilateral medial forebrain bundle. These pathways are a mesodiencephalic system, which terminates in thalamic and hypothalamic nuclei; a mesostriatal projection to the caudate nucleus and putamen; a mesolimbic pathway to the nucleus accumbens, olfactory tubercle, lateral septum, interstitial nucleus of the stria terminalis, amygdala and entorhinal cortex; and mesocortical fibres that project to most cortical areas, particularly the prefrontal, orbitofrontal and cingulate cortices.

Mesencephalic tegmentum

The mesencephalic tegmentum (see Fig. 21.2) is continuous caudally with the pontine tegmentum. At inferior collicular levels, grey matter is restricted to collections of neurones forming the reticular formation and the central (periaqueductal) grey surrounding the cerebral aqueduct. The trochlear nucleus lies in the latter region, near the midline, in a position corresponding to the abducens and hypoglossal nuclei at more caudal levels. It extends through the lower half of the midbrain, just caudal to the oculomotor nucleus and immediately dorsal to the medial longitudinal fasciculus.

The mesencephalic nucleus of the trigeminal nerve occupies a lateral position in the central grey matter. It ascends from the principal sensory nucleus of the trigeminal nerve in the pons to the level of the superior colliculus in the midbrain. Its large ovoid neurones are unipolar, like those in peripheral sensory ganglia. They are arranged in many small groups that extend as curved laminae on the lateral margins of the periaqueductal grey matter; neurones are most numerous caudally.

Apart from these nuclei, the mesencephalic tegmentum contains many other scattered neurones, most of which are included in the reticular formation.

The white matter of the mesencephalic tegmentum contains the majority of the tracts present in the pontine tegmentum. The decussation of the superior cerebellar peduncles is particularly prominent. Numerous fibres that enter the mesencephalic tegmentum arise from the cerebellar nuclei, pass ventromedially, and decussate at the level of the inferior colliculus (see Fig. 21.15). Thereafter, these ascending fibres laterally encapsulate and penetrate the red nucleus, in which some fibres, or their collaterals, terminate (see Fig. 21.16). However, the majority of these fibres ascend to terminate in the nucleus ventralis lateralis of the thalamus. A relatively small number of uncrossed fibres are believed to end in the periaqueductal grey matter and reticular formation, the interstitial nucleus and the nucleus of the posterior commissure. Descending crossed fibres from the decussation of the superior cerebellar peduncle also end in the pontine nuclei, the medullary reticular formation, the nuclei of the inferior olivary complex and, possibly, some cranial nerve motor nuclei.

The medial longitudinal fasciculus adjoins the somatic efferent cell column, dorsal to the decussating superior cerebellar peduncles (see Fig. 21.15). The medial, trigeminal, lateral and spinal lemnisci (anterolateral system) form a curved band dorsomedial to the substantia nigra. Fibres in the medial, trigeminal and spinal lemnisci continue a rostral course to synapse with neurones in the lateral (VPL) and medial (VPM) parts of the ventral posterior thalamic nucleus. Some fibres of the lateral lemniscus end in the nucleus of the inferior colliculus, or send collaterals to it. The remaining lateral lemniscal fibres join with fibres originating in the inferior colliculus and enter the brachium of the inferior colliculus, which carries them to the medial geniculate body.

At the level of the superior colliculus, the tegmentum contains the red nucleus, which extends rostrally into the subthalamic region. The ventromedial central grey matter around the cerebral aqueduct also contains the elongated oculomotor nucleus, which is adjacent to the medial longitudinal fasciculus, and caudally reaches the trochlear nucleus. The oculomotor nucleus is divisible into neuronal groups that are correlated with the motor distribution of the oculomotor nerve. A group of parasympathetic neurones that control the activity of smooth muscle within the eyeball, the Edinger–Westphal preganglionic nucleus, lies dorsal to the oculomotor nucleus. The Edinger–Westphal complex, based on its connections, consists of an Edinger–Westphal preganglionic nucleus projecting only to the ciliary ganglion, and an Edinger–Westphal centrally projecting nucleus that has a variety of targets in the central nervous system but does not target the ciliary ganglion (Kozicz et al 2011).

Red nucleus

The red nucleus is a pink, ovoid mass, approximately 5 mm in diameter and lying dorsomedial to the substantia nigra (see Fig. 21.16). The tint

appears only in fresh material and is caused by a ferric iron pigment in its multipolar neurones. The latter are of varying size, and their relative proportions and arrangement vary between species. In primates the more caudal magnocellular part is decreased, and there is a reciprocal increase in the size of the more rostral parvocellular part. Small multipolar neurones occur in all parts of the nucleus. In humans, the larger neurones of the magnocellular part have been estimated to be as few as 200 in number. The magnocellular element is considered phylogenetically old, which accords with the parvocellular predominance in primates. Rostrally, the red nucleus is poorly demarcated, and it blends into the reticular formation and caudal pole of the interstitial nucleus. It is traversed and surrounded by fascicles of nerve fibres, including many from the oculomotor nucleus.

The principal afferent connections of the red nucleus are corticorubral and cerebellorubral fibres. Uncrossed corticorubral fibres originate from primary somatomotor and somatosensory areas. In animals, the red nucleus receives fibres from the contralateral nucleus interpositus (which corresponds to the human globose and emboliform nuclei) and dentate nucleus, via the superior cerebellar peduncle. It has bilateral, and probably reciprocal, connections with the superior colliculi. In humans, the rubrospinal tract is small and originates from the magnocellular part of the red nucleus. These fibres decussate in the ventral tegmental decussation, ventral to the tectospinal decussation, and ventromedial to the red nucleus. On reaching the grey matter ventral to the inferior cerebellar peduncle, the tract turns caudally to descend adjacent to the lateral lemniscus. It continues descending ventral to the spinal tract of the trigeminal nerve and nucleus throughout the medulla, and enters the upper part of the cervical cord intermingled with fibres of the lateral corticospinal tract (Nathan and Smith 1982). Some efferent axons form a rubrobulbar tract to motor nuclei of the trigeminal, facial, oculomotor, trochlear and abducens nerves. Few fibres reach the cervical cord.

The largest group of efferents from the red nucleus in humans is a descending component of the ipsilateral, central tegmental tract. Initially, this tract lies ventrolateral to the medial longitudinal fasciculus and dorsolateral to both the red nucleus and the decussation of the superior cerebellar peduncles (see Figs 21.10–21.12, 21.15). Most of its fibres arise from the parvocellular part of the red nucleus and pass to the ipsilateral inferior olivary complex in the medulla. Some tract fibres terminate in the brainstem reticular nuclei. Ascending and descending axons from the brainstem reticular formation also run in the central tegmental tract. These axons include dorsal and ventral ascending noradrenergic bundles, a ventral ascending serotonergic bundle, and some fibres of dorsal and ventral ascending cholinergic bundles.

As mentioned elsewhere, lesions of the corticospinal system in humans result in permanent paresis or paralysis. In monkeys, although initially complete, the paralysis disappears and good recovery ensues. The explanation for this interprimate variability in recovery from corticospinal lesions appears to lie in the differential capacity of the rubrospinal system to compensate for loss of corticospinal drive. Monkeys never fully recover from combined lesions of both the corticospinal and rubrospinal tracts, which suggests that the two systems are functionally interrelated in the control of movement. Both encode force, velocity and direction parameters, but the rubrospinal system primarily directs activity both during the terminal phase of a movement and preceding a movement. There is, thus, overlap of activity in the two systems for all parameters during movements of limbs and even of individual digits. The corticospinal system is most active during the learning of new movements, whereas the rubrospinal system is most active during the execution of learned automated movements.

The rubro-olivary projection, which travels in the central tegmental tract, connects the red nucleus indirectly to the contralateral cerebellum via olivocerebellar fibres. The cerebellum is thought to play a role in motor learning, and so the rubro-olivary system could switch the control of movements from the corticospinal to the rubrospinal system for programmed automation. The relative absence of a rubrospinal system in humans could explain the poor recovery of motor function after stroke.

Oculomotor nucleus

The nuclear complex from which the oculomotor nerve arises consists of several groups of large motor neurones (collectively, the oculomotor nucleus; see Fig. 21.16) and smaller preganglionic parasympathetic neurones (the Edinger–Westphal preganglionic nucleus). The motor neurone groups of the oculomotor nucleus innervate, in dorsoventral order, the ipsilateral inferior rectus, inferior oblique and medial rectus. A medially placed column of cells innervates the contralateral superior rectus; the axons from this subnucleus decussate in its caudal part.



The medial rectus subnucleus consists of three anatomically distinct subpopulations. The ventral portion, which contains the largest number of motor neurones, occupies the rostral two-thirds; a subpopulation of smaller motor neurones lies dorsally throughout the rostral two-thirds of the nucleus and innervates the small orbital fibres of the medial rectus; and another subpopulation lies dorsolaterally in the caudal two-thirds of the nucleus.

A median subnucleus of large neurones, the caudal central nucleus, lies at the caudal pole of the oculomotor nucleus adjacent to the superior rectus and medial rectus subnuclei. In non-human primates, approximately 30% of the motor neurones in this subnucleus innervate levator palpebrae superioris bilaterally, which is unique for paired skeletal muscles.

The Edinger–Westphal preganglionic nucleus, composed of small, multipolar, parasympathetic neurones, lies dorsal to the main oculomotor nucleus. Its neurones give rise to axons that travel in the oculomotor nerve and synapse with postganglionic neurones in the ciliary ganglion (Kozicz et al 2011).

Fascicles of axons from these subnuclei course forwards in the midbrain and emerge as the oculomotor nerve in the interpeduncular fossa. The fascicles are most probably arranged from medial to lateral subserving the pupil, inferior rectus, medial rectus, levator palpebrae superioris and superior rectus, and inferior oblique. The human oculomotor nerve contains approximately 15,000 axons.

Afferent inputs to the oculomotor nuclear complex include fibres from the rostral interstitial nucleus of the medial longitudinal fasciculus and the interstitial nucleus of Cajal, both of which are involved in the control of vertical and torsional gaze. Other inputs come, either directly or indirectly, from the nuclei of the posterior commissure, the interstitial nucleus of Cajal, the frontal eye fields, the superior colliculus, the dentate nucleus and other cortical areas. The medial longitudinal fasciculus carries connections from the trochlear, abducens and vestibular nuclei; the medial and lateral vestibular nuclei project to the medial rectus subnucleus.

Afferent inputs to the Edinger–Westphal preganglionic nucleus come primarily from the pretectal nuclei bilaterally, mediating the pupillary light reflex, and from the visual cortex, mediating accommodation.

Trochlear nucleus

The trochlear nucleus lies in the ventromedial periaqueductal grey immediately adjacent to the medial longitudinal fasciculus at the level of the inferior colliculus (see Figs 21.3, 21.15). It is in line with the ventromedial part of the oculomotor nucleus, in the position of the somatic efferent column. The trochlear nucleus is caudal to the oculomotor nucleus and distinguished by the smaller size of its neurones. The afferent inputs to the trochlear nucleus are similar to those described for the oculomotor nucleus. Trochlear efferent fibres pass laterodorsally round the central grey matter, descending caudally medial to the mesencephalic nucleus as they do so, to reach the upper end of the superior medullary velum, where they decussate and emerge lateral to the frenulum and caudal to the inferior colliculus. A few fibres may remain ipsilateral.

Medial longitudinal fasciculus

The medial longitudinal fasciculus (see Figs 21.12, 21.15) is a heavily myelinated composite tract, lying adjacent to the midline, ventral to the periaqueductal and periventricular grey matter. It ascends to the interstitial nucleus of Cajal, which lies in the lateral wall of the third ventricle, just above the cerebral aqueduct. The fasciculus retains its position relative to the central grey matter through the midbrain, pons and upper medulla, but is displaced ventrally by the motor (pyramidal) decussation containing corticospinal fibres.

The medial longitudinal fasciculus interconnects the oculomotor, trochlear, abducens, Edinger–Westphal preganglionic, vestibular, reticular and spinal accessory nuclei, coordinating conjugate eye movements and associated movements of the head and neck. Lesions cause internuclear ophthalmoplegia. All four vestibular nuclei contribute ascending fibres. Those from the superior nucleus remain uncrossed, while the others are partly crossed. Some fibres reach the interstitial and posterior commissural nuclei, and some decussate to the contralateral nuclei. Descending axons, from the medial vestibular nuclei and perhaps the lateral and inferior nuclei, partially decussate and descend in the fasciculus to form the medial vestibulospinal tract, which travels in the medial longitudinal fasciculus into the ventral funiculus of the spinal cord (see Ch. 20 and Fig. 20.18). Fibres join the fasciculus from the dorsal trapezoid, lateral lemniscal and posterior commissural nuclei, which means that both the cochlear and vestibular components of the vestibulocochlear nerve may influence movements of the eyes and head via the medial longitudinal fasciculus. Some vestibular fibres may ascend in the medial longitudinal fasciculus as far as the thalamus.

Tectum



Inferior colliculus

The inferior colliculus (see Fig. 21.15) is part of the ascending auditory pathway and is the principal site of termination of the lateral lemniscus. It has a central, ovoid, main nucleus, which is lateral to the periaqueductal grey matter. It is surrounded by a lamina of nerve fibres, many from the lateral lemniscus, which terminate in it. The central nucleus

has dorsomedial and ventrolateral zones, which are covered by a dorsal cortex. In humans, the cortex has four cytoarchitectonic layers: layer I contains small neurones with flattened radial dendritic fields; layer II, medium-sized neurones with ovoid dendritic fields aligned parallel with the collicular surface; layer III, medium-sized neurones with spherical dendritic fields; and layer IV, large neurones with variably shaped dendritic fields.

Most efferent fibres from the inferior colliculus travel via the inferior brachium to the ipsilateral medial geniculate nucleus. Lemniscal fibres relay only in the central nucleus, and some pass without relay to the medial geniculate nucleus. In humans, the ventral division of the medial geniculate nucleus receives a topographic projection from the central nucleus and the dorsal division receives a similar projection from the dorsal cortex. Some colliculogeniculate fibres do not relay in the geniculate nucleus but continue, with those that do, via the auditory radiation to the auditory cortex. A descending projection from the auditory cortex reaches the inferior colliculus via the medial geniculate nucleus. Some fibres may traverse this projection without relay. This descending path may produce effects at levels from the medial geniculate nucleus downwards, and it probably links with efferent cochlear fibres, through the superior olivary and cochlear nuclei.

Superior colliculus

The superior colliculus is an obviously laminated structure. At successive depths from the external surface, it may be divided into seven layers termed zonal, superficial grey, optic, intermediate grey, deep grey, deep white and periventricular strata, composed alternately of neuronal somata or their processes. The zonal layer consists chiefly of myelinated and non-myelinated fibres from the occipital cortex (areas 17, 18 and 19), which arrive as the external corticotectal tract. It also contains a few small neurones, which are horizontally arrayed. The superficial grey layer (stratum cinereum) forms a crescentic lamina over the deeper layers and contains many small multipolar interneurones, on which cortical fibres synapse. The optic layer consists partly of fibres from the optic tract. As they terminate, they permeate the entire anterior–posterior extent of the superficial layers with numerous collateral branches. This arrangement provides a retinotopic map of the contralateral visual field, in which the fovea is represented anterolaterally. Retinal axons terminate in clusters from specific retinotectal neurones and as collaterals of retinogeniculate fibres. The layer also contains some large multipolar neurones. Efferent fibres to the retina are said to start in this layer.

The intermediate grey and white layers collectively constitute the main receptive zone. The main afferent input is the medial corticotectal path from layer V neurones of the ipsilateral occipital cortex (area 18), and from other neocortical areas that are concerned with ocular following movements. Afferent fibres are also received from the contralateral spinal cord (via spinotectal fibres in the anterolateral system), the inferior colliculus, and the locus coeruleus and raphe nuclei (from noradrenergic and serotonergic neurones). The deep grey and deep white layers adjacent to the periaqueductal grey matter are collectively called the parabrachial nucleus. They contain neurones whose dendrites extend into the optic layer, and whose axons form many of the collicular efferents.

The superior colliculus receives afferents from many sources including the retina, spinal cord, inferior colliculus and occipital and temporal cortices. The first three of these pathways convey visual, tactile and probably thermal, pain and auditory impulses. Collicular efferents pass to the retina, lateral geniculate nucleus, pretectum, parabrachial nucleus, the inferior, medial and lateral pulvinar, and to numerous sites in the brainstem and spinal cord. Fibres passing from the pulvinar are relayed to primary and secondary visual cortices and form an extrageniculate retinocortical pathway for visual orientation and attention.

The tectospinal and tectobulbar tracts start from neurones in the superior colliculi. They sweep ventrally around the central grey matter to decussate ventral to the oculomotor nuclei and medial longitudinal fasciculi as part of the dorsal tegmental decussation. The tectospinal tract descends ventral to the medial longitudinal fasciculus throughout the brainstem, diverges ventrolaterally at the level of the motor decussation, and moves medially into the ventral funiculus in association with the medial longitudinal fasciculus. Tectospinal fibres descend to cervical segments. The tectobulbar tract, mainly crossed, descends near the tectospinal tract, and ends in the pontine nuclei and motor nuclei of the cranial nerves, particularly those innervating the oculogyric muscles. It subserves reflex ocular movements. Other tectotegmental fibres reach various tegmental reticular nuclei in the ipsilateral mesencephalic and contralateral pontomedullary reticular formation (gigantocellular reticular, caudal pontine reticular, oral pontine reticular

The central nucleus contains indistinct laminae. Bands of cells with disc-shaped or stellate dendritic fields orthogonally span the fibre layers in which the terminals of lateral lemniscal fibres ramify. The neurones are sharply tuned to frequency, and the laminae may represent the structural basis of tonal discrimination. Experimental studies have found cells driven by low frequencies in the dorsal laminae, and others driven by high frequencies in the ventral laminae. Neurones are broadly frequency-tuned in the dorsal cortex and lateral nucleus.

Inferior collicular projections to the brainstem and spinal cord appear to traverse the superior colliculi before they descend. In this way they connect with the origins of the tectospinal and tectotegmental tracts. These projections are relatively small and probably mediate reflex turning of the head and eyes in response to sounds.

In experimental animals, lesions of either the inferior colliculus or its brachium produce defects in tonal discrimination, sound localization and auditory reflexes. The effects of such lesions are poorly documented in humans.

nuclei), and the substantia nigra and red nucleus. Tectopontine fibres, which probably descend with the tectospinal tract, terminate in dorso-lateral pontine nuclei, with a relay to the cerebellum.

Pretectal nucleus

The pretectal nuclei, of which the olivary pretectal nucleus is one, is a poorly defined mass of neurones at the junction of the mesencephalon and diencephalon. It extends from a position dorsolateral to the posterior commissure caudally towards the superior colliculus, with which it is partly continuous. It receives fibres from the visual cortex via the superior quadrigeminal brachium, from the retina via the lateral root of the optic tract, and from the superior colliculus. Its efferent fibres reach the Edinger–Westphal preganglionic nuclei bilaterally. Those that decussate pass ventral to the aqueduct or through the posterior commissure. In this way, the sphincter pupillae contract in both eyes in response to impulses from either eye. This bilateral light reflex may not be the sole activity of the pretectal nucleus. Some of its efferents project to the pulvinar and deep laminae of the superior colliculus, and provide another extrageniculate path to the cerebral cortex.



BRAINSTEM RETICULAR FORMATION

The brainstem contains extensive fields of intermingled neuronal cell bodies and nerve fibres, which are collectively termed the reticular formation (Haines 2013, Nieuwenhuys et al 2008, Olszewski and Baxter 1954). The reticular regions are often regarded as phylogenetically ancient, representing a primitive nerve network on which more anatomically organized, functionally selective connections have developed during evolution.

Reticular regions tend to be ill-defined collections of neurones with diffuse connections. Their conduction paths are difficult to define, often complex and polysynaptic, and they have ascending and descending components that are crossed, uncrossed and sometimes bilateral. Their components subserve somatic and visceral functions. They also include some distinct and important cell groups, which are distinguished on the basis of their connections and neurotransmitter substances. These include dopaminergic and noradrenergic neurones (group A), serotonergic (group B), adrenergic (group C) and cholinergic (group Ch) neurones (Dahlström and Fuxe 1964, Dahlström and Fuxe 1965, Nieuwenhuys 1985, Siegel et al 1999).

Reticular neurones have long dendrites that spread along the long axis of the brainstem. The dendrites may spread into 50% of the cross-sectional area of their half of the brainstem, and they are intersected by, and may synapse with, a complex of ascending and descending fibres. Many axons of reticular neurones ascend or descend, or bifurcate in both directions. They often travel far, perhaps through the whole brainstem and beyond.

Multitudes of afferent fibres converging on individual neurones, and the myriad destinations of efferent fibres provide the structural basis for considering the reticular formation to be 'polymodal', 'diffuse' or 'non-specific' in function.

A contrasting dendritic form is also found, in which dendrites are short, sinuous or curved, and branch profusely at the perimeter of a nuclear group, defining a boundary between it and its environs. In different zones, the proportion of different sizes of neuronal somata varies. Some regions contain only small to intermediate multipolar cells ('parvocellular' regions). However, there are a few areas where these mingle with large multipolar neurones in 'gigantocellular' or 'magnocellular' nuclei.

In general terms, the reticular formation is a continuous core that traverses the whole brainstem, and is continuous below with the reticular intermediate spinal grey. It is divisible, on the basis of cytoarchitectonic, chemoarchitectonic and functional criteria, into three bilateral longitudinal columns: median; medial, containing mostly large reticular neurones; and lateral, containing mostly small to intermediate neurones (Fig. 21.18).

MEDIAN COLUMN OF RETICULAR NUCLEI



The median column of reticular nuclei extends throughout the medulla, pons and midbrain and contains neurones that are largely aggregated in bilateral, vertical sheets, located immediately adjacent to the midline and occupying the paramedian zones. Collectively they are called the nuclei of the raphe, or raphe nuclei. Many neurones in raphe nuclei are serotonergic and are grouped into nine clusters, B1–9 (see Fig. 21.17) (Dahlström and Fuxe 1964, Dahlström and Fuxe 1965, Nieuwenhuys 1985, Hornung 2003). The raphe pallidus nucleus (B1) and associated raphe obscurus nucleus (B2) lie in the upper two-thirds of the medulla and cross the pontomedullary junction. The raphe magnus nucleus, corresponding to many B3 neurones, minimally overlaps with B1 and

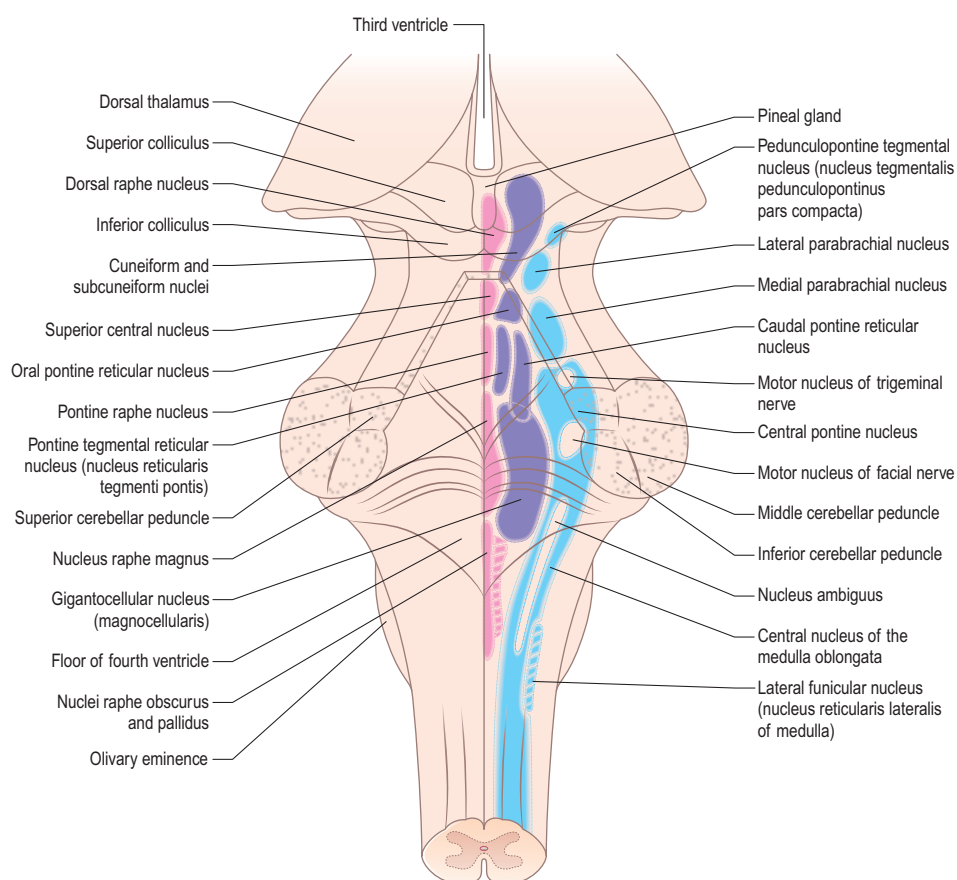


Fig. 21.18 The dorsal aspect of the brainstem showing the approximate location of nuclei of the reticular formation. Nuclei of the median and paramedian nuclear column, pink; medial column nuclei, purple; lateral column nuclei, blue.

A tecto-olivary projection, from deeper collicular laminae to the upper third of the medial accessory olivary nucleus, exists in primates. It is crossed and links with the posterior vermis.

In animals, central collicular stimulation produces contralateral head movement as well as movements involving the eyes, trunk and limbs, which implicates the superior colliculus in the integration of vision and body movement.

As an example, a bifurcating axon from a cell in the magnocellular medullary nucleus may project rostrally into the upper medulla, pons, midbrain tegmentum, subthalamus, hypothalamus, dorsal thalamus, septum, limbic system and neocortex, while its descending branch innervates the reticular core of the lower medulla and may reach the cervical spinal intermediate grey of laminae V and VI. Many reticular neurones have unidirectional, shorter axons that synapse with the radiating dendrites of numerous other neurones *en route*, and give off collaterals that synapse with cells in 'specific' brainstem nuclei or cortical formations, such as the cerebellum.

B2, and ascends into the caudal pons. Above it is the pontine raphe nucleus, which is formed by cell group B5. Also located in the pons is the central superior raphe nucleus, which contains parts of cell groups B6 and B8. The dorsal (rostral) raphe nucleus, approximating to cell group B7, extends through much of the midbrain. B4 and B9 are small cell groups located in the medulla and midbrain, respectively.

Axons originating from the serotonergic raphe neurones ramify extensively throughout the CNS.

All raphe nuclei provide descending serotonergic projections, which terminate in the brainstem and spinal cord. Brainstem connections are multiple and complex. For example, the dorsal raphe nucleus, in addition to sending a large number of fibres to the locus coeruleus, projects to the dorsal tegmental nucleus and most of the rhombencephalic reticular formation, together with the central superior, pontine raphe and raphe magnus nuclei.

Raphe-spinal serotonergic axons originate mainly from neurones in the raphe magnus, pallidus and obscurus nuclei. They project as ventral, dorsal and intermediate spinal tracts in the ventral and lateral funiculi, and terminate respectively in the ventral horns and laminae I, II and V of the dorsal horns of all segments, and in the thoracolumbar intermediolateral sympathetic and sacral parasympathetic preganglionic cell columns. The dorsal raphe spinal projections function as a pain control pathway that descends from a mesencephalic pain control centre located in the periaqueductal grey matter, dorsal raphe and cuneiform nuclei. The intermediate raphe-spinal projection is inhibitory and, in part, modulates central sympathetic control of cardiovascular function. The ventral raphe-spinal system excites ventral horn cells and could function to enhance motor responses to nociceptive stimuli.

Principally, the mesencephalic serotonergic raphe system is reciprocally interconnected rostrally with the limbic system, septum, prefrontal cortex and hypothalamus. Efferents ascend and form a large ventral and a diminutive dorsal pathway.

Major afferents to the mesencephalic raphe nuclei include those from the interpeduncular nucleus, linking the limbic and serotonergic systems; from the lateral habenular nucleus, linking the septum, preoptic hypothalamus and prefrontal cortex via the fasciculus retroflexus and the medial forebrain bundle; and from the pontine central grey matter.

The ascending raphe system probably functions to moderate forebrain activities, particularly limbic, septal and hypothalamic activities. A degree of region-specific connectivity suggests that it exerts precise, as well as tonal, control.

MEDIAL COLUMN OF RETICULAR NUCLEI

The medial column of reticular nuclei is composed predominantly of neurones of medium size, although very large neurones are found in some regions, and most have processes orientated in the transverse plane. In the lower medulla the column is indistinct, and is perhaps represented by a thin lamina lateral to the raphe nuclei. However, in the upper medulla it expands into the medullary gigantocellular (magnocellular) nucleus, which lies ventrolateral to the hypoglossal nucleus, ventral to the vagal nuclei and dorsal to the inferior olivary complex. Ascending further, the column continues as the pontine gigantocellular (magnocellular) nucleus, which lies medially in the tegmentum. Its neurones suddenly diminish in size to form, in rostral order, the almost coextensive caudal and oral pontine tegmental reticular nuclei. It then expands into the cuneiform nucleus and subcuneiform nucleus, before fading away in the midbrain tegmentum (Haines 2013, Nieuwenhuys et al 2008).

Axons of medial reticular column neurones form a multisynaptic ascending and descending system within the column, and ultimately enter the spinal cord and diencephalon. Descending fibres form pontine (medial) reticulospinal and medullary (lateral) reticulospinal tracts. Pontine reticulospinal axons arise from neurones in the caudal and oral parts of the pontine reticular nucleus, descend largely uncrossed in the ventral spinal funiculus, and terminate in spinal cord laminae VII, VIII and IX. Medullary reticulospinal axons descend bilaterally with an ipsilateral preponderance to end in laminae VII, VIII, IX and X, and ipsilaterally to end in laminae IV, V and VI. The system modulates spinal motor function and segmental nociceptive input.

Afferent components to the medial reticular nuclear column include the spinoreticular projection and collaterals of centrally projecting spinal trigeminal, vestibular and cochlear fibres. Spinoreticular fibres, part of the anterolateral system, arise from neurones in the intermediate grey matter of the spinal cord. They decussate in the ventral white commissure, ascend in the ventrolateral funiculus, and terminate not only

at all levels of the medial column of reticular nuclei but also in the intralaminar nuclei of the thalamus.

LATERAL COLUMN OF RETICULAR NUCLEI

The lateral column of reticular nuclei contains six nuclear groups. These are the parvocellular reticular area, superficial ventrolateral reticular area, lateral pontine tegmental noradrenergic cell groups A1, A2, A4–A7 (A3 is absent in primates), adrenergic cell groups C1–C3 and cholinergic cell groups Ch5–Ch6 (Nieuwenhuys 1985, Siegel et al 1999). Cell groups Ch1–Ch4 are located in the basal forebrain. The column descends through the lower two-thirds of the lateral pontine tegmentum and upper medulla, where it lies between the gigantocellular nucleus medially, and the sensory nuclei of the trigeminal nerve laterally. It continues caudally and expands to form most of the reticular formation lateral to the raphe nuclei. It abuts the superficial ventrolateral reticular area, nucleus solitarius, nucleus ambiguus and vagal nucleus, and there contains the adrenergic cell group C2 and the noradrenergic group A2.

The lateral paragigantocellular nucleus lies at the rostral pole of the diffuse superficial ventrolateral reticular area (at the level of the facial nucleus). The zone extends caudally as the nucleus retroambiguus and descends into the spinal cord. It contains noradrenergic cell groups A1, A2, A4 and A5, and the adrenergic cell group C1. The ventrolateral reticular area is involved in cardiovascular, respiratory, vasoreceptor and chemoreceptor reflexes, and in the modulation of nociception. The A2 or noradrenergic dorsal medullary cell group lies in the nucleus of the tractus solitarius, vagal nucleus and adjoining parvocellular reticular area. Adrenergic group C1 lies rostral to the A2 cell group. Noradrenergic cell group A4 extends into the lateral pontine tegmentum, along the subependymal surface of the superior cerebellar peduncle. Noradrenergic group A5 forms part of the paragigantocellular nucleus in the caudolateral pontine tegmentum. Noradrenergic cell group A5 and adrenergic cell group C1 probably function as centres of vasomotor control. The entire region is subdivided into functional areas on the basis of experiments in animals, in which vasoconstrictor, cardio-accelerator, depressor, inspiratory, expiratory and sudomotor responses have been elicited.

The lateral pontine tegmental reticular grey matter is related to the superior cerebellar peduncle and forms the medial and lateral parabrachial nuclei and the ventral Kölliker–Fuse nucleus, a pneumotaxic centre. The locus coeruleus (noradrenergic cell group A6), area subcoeruleus, noradrenergic cell group A7 and cholinergic group Ch5 in the pedunculopontine tegmental nucleus (nucleus tegmentalis pedunculopontinus) are all located in the lateral pontine and mesencephalic tegmental reticular zones. A4 is sometimes considered part of the coeruleus complex. The mesencephalic group Ch5 is continuous caudally with cell group Ch6 in the pontine central grey matter.

Cell group A6 (also parts of A4) contains all the noradrenergic cells in the central region of the locus coeruleus. Group A6 has ventral (nucleus subcoeruleus, A6sc), rostral and caudolateral extensions (A6cg), the latter merging with the A4 group. The locus coeruleus probably functions as an attention centre, focusing neural functions to prevailing needs. The noradrenergic A7 group occupies the rostroventral part of the pontine tegmentum and is continuous with groups A5 and A1 through the lateral rhombencephalic tegmentum. The A7, A5, A1 complex is also connected by noradrenergic cell clusters with group A2, caudally, and group A6, rostrally. The A5 and A7 groups lie mainly within the medial parabrachial and Kölliker–Fuse nuclei. Reticular neurones in the lateral pontine tegmental reticular area, like those of the ventrolateral zone, function to regulate respiratory, cardiovascular and gastrointestinal activity. Two micturition centres are located in the dorsomedial and ventrolateral parts of the lateral pontine tegmentum.

The connections of the lateral column reticular nuclei are complex. The short ascending and descending axons of the parvocellular reticular area constitute medullary reflex pathways, which connect all branchiomotor nuclei and the hypoglossal nucleus with central afferent cranial nerve complexes through a propriobulbar system. This area also receives descending afferents from the contralateral motor cortex via the corticotegmental tract, and from the contralateral red nucleus via the rubrospinal tract. The longitudinal catecholamine bundle passes through the parvocellular reticular formation.

The superficial ventrolateral reticular area receives its principal input from the nucleus solitarius, and subserves cardiovascular, baroreceptor, chemoreceptor and respiratory reflexes; it also receives some input from the spinal cord, insular cortex and amygdala. Reticulospinal fibres from this region terminate bilaterally on sympathetic preganglionic neurones

Although many of these fibres are diffusely distributed, some have more specific connections. For example, whereas the central superior raphe nucleus projects divergently to all areas of the cerebral cortex, different neurones in the dorsal raphe nucleus project specifically not only to circumscribed regions of the frontal, parietal and occipital cortices, but also to functionally related regions of the cerebellar cortex. Similarly, the caudate nucleus and putamen receive a preferential input from the dorsal raphe nucleus, whereas the hippocampus, septum and hypothalamus are innervated mainly by cells in the central superior mesencephalic raphe nucleus.

Both originate from neurones in the dorsal and central superior raphe nuclei. The raphe magnus nucleus also contributes to the dorsal ascending serotonergic pathway, which is at first incorporated into the dorsal longitudinal fasciculus (of Schütz). A few fibres terminate in the central mesencephalic grey matter and posterior hypothalamus, but most continue into the medial forebrain bundle and merge with the axons of the ventral pathway, which are distributed to the same targets. The fibres of the ventral ascending serotonergic pathway exit the ventral aspect of the mesencephalic raphe nuclei, and then course rostrally through the ventral tegmentum from where fibres pass to the ventral tegmental area, substantia nigra and interpeduncular nucleus. A large number of fibres then enter the fasciculus retroflexus (habenulo-interpeduncular tract) and run rostrally to innervate the habenular nucleus, intralaminar, midline, anterior, ventral and lateral dorsal thalamic nuclei, and the lateral geniculate body. The ventral ascending

serotonergic pathway enters the medial forebrain bundle in the lateral hypothalamic area and splits to pass medially and laterally. The fibres in the medial tract terminate in the mammillary body, dorsomedial, ventromedial, infundibular, anterior and lateral hypothalamic, medial and lateral preoptic and suprachiasmatic nuclei. Those in the lateral tract take the ansa peduncularis–ventral amygdalofugal path to the amygdala, striatum and neocortex. The medial forebrain bundle carries the remaining ventral ascending serotonergic axons into the medullary stria, stria terminalis, fornix, diagonal band, external capsule, cingulate fasciculus and medial olfactory stria, to terminate in all the structures that these systems interconnect.

Three areas of the medial reticular zone receive particularly high densities of terminations. These are the combined caudal and rostral ends of the gigantocellular and central nuclei, respectively, and the caudal pontine reticular nucleus and the pontine tegmentum. Retinotectal and tectoreticular fibres relay visual information and the medial forebrain bundle transmits olfactory impulses.

Efferents from the medial column of reticular nuclei project through a multisynaptic pathway within the column to the thalamus. Areas of maximal termination of spinoreticular fibres also project directly to the intralaminar thalamic nuclei. The multisynaptic pathway is integrated into the lateral column of reticular nuclei with cholinergic neurones in the lateral pontine tegmentum. The intralaminar thalamic nuclei project directly to the striatum and neocortex.

in the thoracic spinal cord. Fibres from the pneumotaxic centre project to an inspiratory centre in the ventrolateral part of the nucleus solitarius, and a mixed expiratory–inspiratory centre in the superficial ventrolateral reticular area. Inspiratory neurones in both centres monosynaptically project to the phrenic and intercostal motor neurones. Axons of expiratory neurones terminate on lower motor neurones that innervate intercostal and abdominal musculature.

Reticulospinal fibres descend from the lateral pontine tegmentum. A mainly ipsilateral subcoeruleospinal pathway is distributed to all spinal segments of the cord through the lateral spinal funiculus. Crossed pontine reticulospinal fibres descend from the ventrolateral pontine tegmentum, decussate in the rostral pons and occupy the contralateral dorsolateral spinal funiculus. They terminate in laminae I, II, V and VI of all spinal segments of the cord. Fibres from the pneumotaxic centre innervate the phrenic nucleus and T1–T3 sympathetic preganglionic neurones bilaterally through this projection system.

Bilateral projections from the micturition centres travel in the lateral spinal funiculus. They terminate on preganglionic parasympathetic neurones in the sacral cord (which innervate the detrusor muscle in the urinary bladder), and on neurones in the nucleus of Onuf (which innervate the musculature of the pelvic floor and the anal and urethral sphincters).

Descending fibres of the A6 noradrenergic neurones of the locus coeruleus project into the longitudinal dorsal fasciculus (as the caudal limb of the dorsal periventricular pathway), and into the caudal limb of the dorsal noradrenergic bundle (as part of the longitudinal catecholamine bundle). In this way they innervate, mainly ipsilaterally, all other rhombencephalic reticular areas, principal and spinal nuclei of the trigeminal nerve, pontine nuclei, cochlear nuclei, nuclei of the lateral lemniscus, and bilaterally, all spinal preganglionic autonomic neurones and the ventral region of the dorsal horn in all segments of the spinal cord.

Most ascending fibres from the locus coeruleus pass in the dorsal noradrenergic (or tegmental) bundle; others run either in the rostral limb of the dorsal periventricular pathway or in the superior cerebellar peduncle. The latter fibres terminate on the cerebellar nuclei. The dorsal noradrenergic bundle is large and runs through the ventrolateral periaqueductal grey matter to join the medial forebrain bundle in the hypothalamus, from where fibres continue forwards to innervate all rostral areas of the brain. The pathway contains efferent and afferent axons that reciprocally connect the locus coeruleus with adjacent structures along its course, e.g. central mesencephalic grey matter, dorsal raphe nucleus, superior and inferior colliculi, interpeduncular nucleus, epithalamus, dorsal thalamus, habenular nuclei, amygdala, septum, olfactory bulb and anterior olfactory nucleus, the entire hippocampal formation and neocortex.

Pedunculopontine nucleus

The pedunculopontine nucleus (nucleus tegmentalis pedunculopontinus; PPN) lies in the dorsolateral part of the pontomesencephalic tegmentum. Despite some terminological confusion, most authors agree

that the pedunculopontine nucleus is bounded laterally by the medial lemniscus, medially by the superior cerebellar peduncle and its decussation, dorsally by the pontine cuneiform and subcuneiform nuclei and ventrally by the pontine reticular formation. Rostrally, its ventral border contacts the dorsomedial aspect of the substantia nigra and it is separated from the subthalamic nucleus by the retrorubral field. Its most caudal pole is adjacent to the locus coeruleus (Pahapill and Lozano 2000).

The pedunculopontine nucleus has been subdivided into two territories based on their cytoarchitectonic and neurochemical characteristics: namely, a caudal pars compacta consisting mainly of cholinergic neurones, and a rostral pars dissipata consisting of approximately equal numbers of cholinergic and glutamatergic neurones scattered from the mid-mesencephalic to mid-pontine levels (Hamani et al 2007). The pedunculopontine nucleus is encircled by the mesencephalic locomotor region (MLR), a region that also includes the cuneiform and subcuneiform nuclei. Based on experimental studies in animals, where stimulation of the mesencephalic locomotor region elicits locomotion, the pedunculopontine nucleus has been implicated in the initiation and modulation of gait and other stereotyped movements. Functionally, the pedunculopontine nucleus is associated with the reticular activating system via its cholinergic and glutaminergic connections with intralaminar thalamic nuclei. However, its most important and complex connections are reciprocal pathways with the basal ganglia, especially the internal globus pallidus and the substantia nigra; these pathways are described in detail in Chapter 24.

BRAINSTEM LESIONS

GENERAL CONCEPTS

Brainstem lesions (Figs 21.19–21.20) may arise as a result of extrinsic compression of the brainstem by space-occupying tumours (e.g. meningioma, vestibular schwannoma, metastatic carcinoma) or may be caused by intrinsic disease (e.g. glioma, demyelination, stroke). The clinical syndrome is generally characterized by: ipsilateral cranial nerve deficits, and a contralateral hemiplegia (corticospinal involvement) and/or contralateral hemianaesthesia (anterolateral system, dorsal column–medial lemniscus). Brainstem lesions may also result in cerebellar signs (damage to cerebellar afferents or efferents) or in symptoms reflecting small defects (nystagmus, internuclear ophthalmoplegia) (Haines 2013, Posner et al 2007).

In cases of brainstem lesions, the cranial nerve deficit is the best localizing sign; it specifies the side of the lesion (cranial nerves receive input from and project to the ipsilateral side) and the level within the brainstem. Midbrain lesions (cranial nerves III, IV) cause ophthalmoplegia, pupillary dilation and ptosis (oculomotor nerve palsy), and impaired upward gaze (e.g. due to a pinealoma). Mid-pontine lesions (cranial nerve V) may result in a loss of sensation on the face and in the oral cavity and weakness of the masticatory muscles. Tumours or

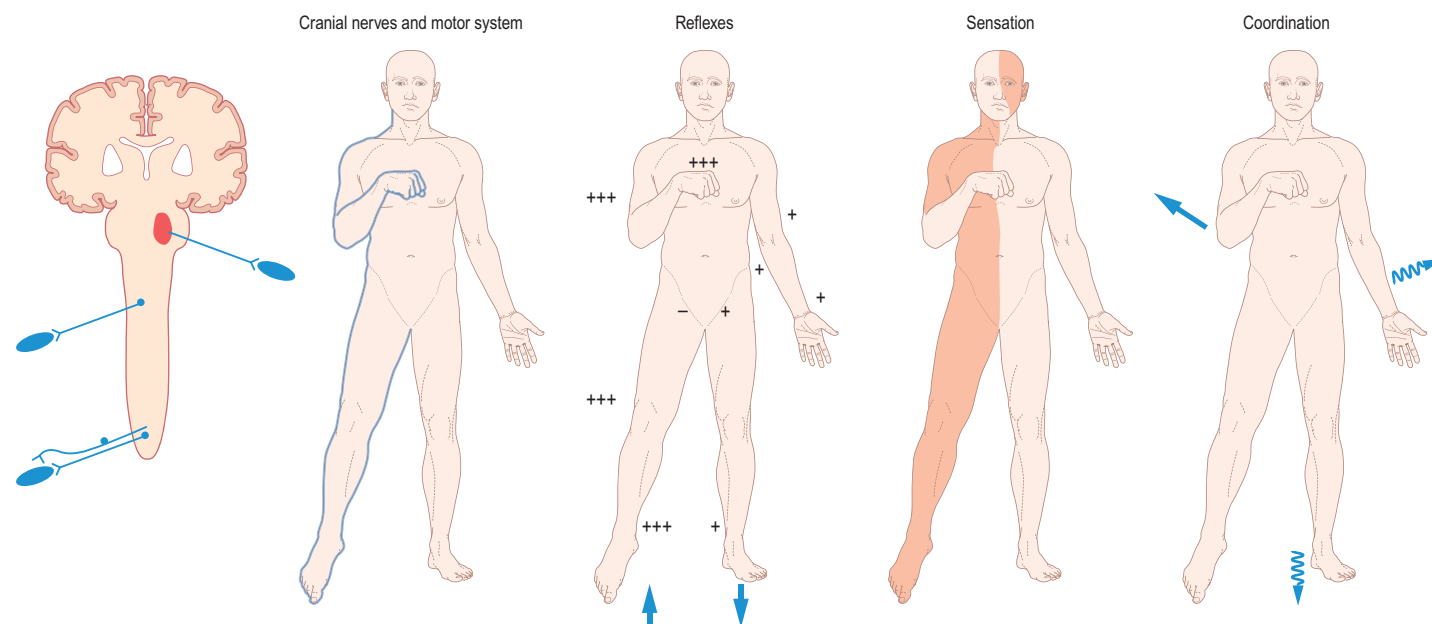


Fig. 21.19 Brainstem lesions. (With permission from Crossman AR, Neary D 2010 Neuroanatomy, 4th edn. Edinburgh: Churchill Livingstone.)

The superficial ventrolateral area is also the seat of the 'visceral alerting response'. Fibres from the hypothalamus, periaqueductal grey matter and midbrain tegmentum mediate increased respiratory activity, raised blood pressure, tachycardia, vasodilation in skeletal muscle and renal and gastrointestinal vasoconstriction. Ascending efferents from the superficial ventrolateral area synapse on neurones of the supraoptic and paraventricular hypothalamic nuclei. Excitation of these neurones causes release of vasopressin from the neurohypophysis. Medullary noradrenergic cell groups A1 and A2 also innervate (directly and indirectly) the median eminence, and control the release of growth hormone, luteinizing hormone and adrenocorticotrophic hormone.

The lateral pontine tegmentum, particularly the parabrachial region, is reciprocally connected to the insular cortex. It shares reciprocal projections with the amygdala through the ventral amygdalofugal pathway, medial forebrain bundle and central tegmental tract, and with hypothalamic, median preoptic and paraventricular nuclei, which preferentially project to the lateral parabrachial nucleus and the micturition centres. It also shares reciprocal bulbar projections, many from the pneumotaxic centre, with the nucleus solitarius and superficial ventrolateral reticular area.

Other axons that contribute to the longitudinal catecholamine bundle originate from cell groups C1, A1, A2, A5 and A7. The main projection is a descending one from cell groups C1 and A5, which are sudomotor neural control centres and innervate preganglionic sympathetic neurones.

Fibres from the locus coeruleus that travel in the rostral limb of the dorsal periventricular pathway ascend in the ventromedial periaqueductal grey matter adjacent to the longitudinal dorsal fasciculus and terminate in the parvocellular part of the paraventricular nucleus in the hypothalamus.

The functions of the locus coeruleus and related tegmental noradrenergic cell groups are incompletely understood. The diversity of their rostral and caudal projections suggests a widespread role in central

processing. In animals, firing rates of locus coeruleus neurones peak during wakefulness and decrease during sleep – they cease almost completely during rapid eye movement (REM) sleep. During wakefulness, firing rates are augmented when novel stimuli are presented. The locus coeruleus may, therefore, function to control the level of attentiveness. Other functions that have been ascribed to the locus coeruleus include control of the wake–sleep cycle, regulation of blood flow, and maintenance of synaptic plasticity.

The A1, A2, A5 and A7 noradrenergic cell groups project rostrally, mainly through the central tegmental tract. Their axons constitute a major longitudinal catecholamine pathway that continues through the medial forebrain bundle and ends in the amygdala, lateral septal nucleus, bed nucleus of the stria terminalis, nucleus of the diagonal band and the hypothalamus. The ascending dorsal periventricular pathway contains a few non-coerulean noradrenergic fibres, which terminate in the periventricular region of the thalamus.

Projections arising in the rhombencephalon receive contributions from the diffusely organized dorsal medullary and lateral tegmental noradrenergic cell groups. These interconnect cranial nerve nuclei and other reticular cell groups, particularly those of the vagus, facial and trigeminal nerves, and the rhombencephalic raphe and parabrachial nuclei.

Precerebellar nuclei are those that send axons to the cerebellar cortex and nuclei. The three related to the reticular formation are the lateral and paramedian reticular nuclei, and the reticulotegmental nucleus; these are involved in the relay of spinal information into primarily the vermis and paravermal regions of the ipsilateral cerebellar hemisphere. They receive inputs from the contralateral primary motor and sensory neocortices, and the ipsilateral cerebellar and vestibular nuclei and spinal cord (the latter through the ascending spinoreticular pathway). This system augments the dorsal and ventral spinocerebellar, cuneocerebellar, accessory cuneocerebellar and trigeminocerebellar tracts.

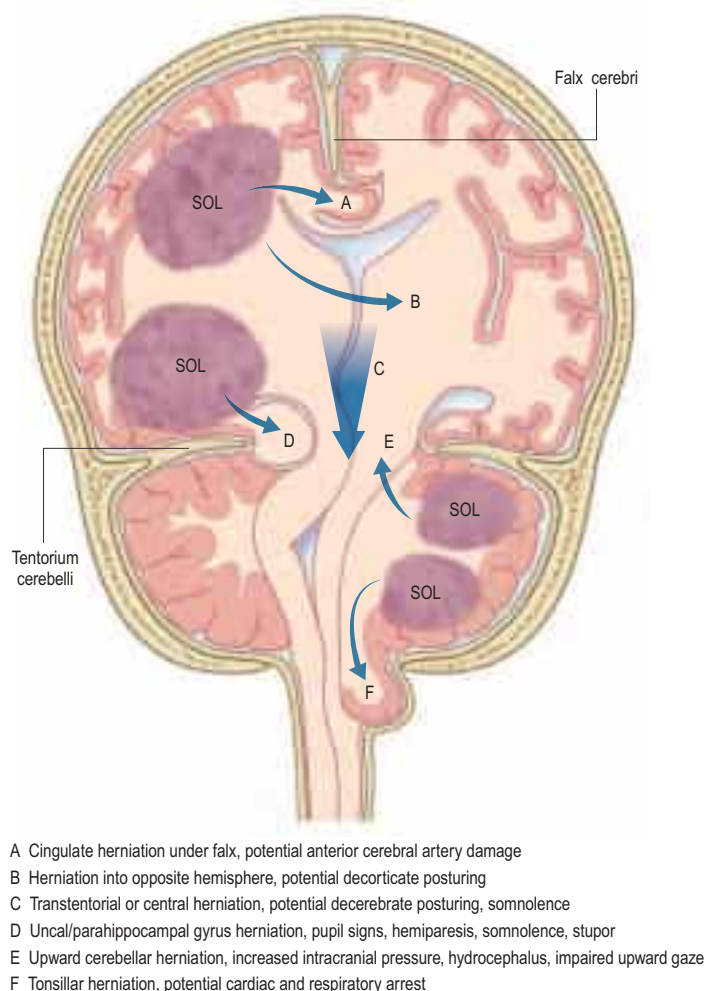


Fig. 21.20 Space-occupying lesions (SOL).

lesions at the level of the pontomedullary junction (cranial nerves VI, VII, VIII) may produce weakness of the lateral rectus muscle, a loss of facial sensation, and deafness, vertigo or tinnitus. Medullary lesions (cranial nerves IX, X, XII and traversing fibres of XI) may cause a 'bulbar palsy': dysarthria, dysphagia and dysphonia, with wasting of the hemitongue, and deviation, weakness and wasting of sternocleidomastoid and trapezius. All are usually accompanied by signs or symptoms of long tract involvement (motor, sensory or both).

Herniation syndromes

Damage to the brainstem resulting in characteristic deficits is a major element of what are called herniation syndromes. An increase in

intracranial pressure, signalled by headache, nausea and papilloedema in one compartment (tumour, haemorrhage, abscess, metabolic disorder), usually results in an extrusion of CNS structures from a location of higher pressure to an adjacent location of lower pressure: the brain herniates. All herniation syndromes, directly or indirectly, will compromise brainstem viability (Posner et al 2007, Haines 2015).

An expanding lesion in parietofrontal areas, a supratentorial position, may result in herniation of structures such as the cingulate gyrus underneath the edge of the falx cerebri (see Fig. 21.20A). This may initially be a 'silent' event (no signs or symptoms) or may present as lower-extremity weakness related to compression of the anterior cerebral artery (ipsilateral, contralateral or bilateral) and decreased levels of consciousness.

Hemisphere lesions may enlarge, impinge into the opposite side of the brain, and produce a characteristic series of deficits (see Fig. 21.20B). These include decrease in consciousness indicating damage to the thalamic relays of the reticular activating system, generally increased muscle tone, a change in respiratory patterns, and altered pupils and eye movements. These patients may become decorticate: lower extremities extended, upper extremities flexed.

As a cerebral mass compromises the available supratentorial space and the clinical picture deteriorates, the herniation traverses the tentorial notch as a transtentorial (or central) herniation (see Fig. 21.20C). These patients have a significant decrease in the level of consciousness (some may be stuporous or in coma); breathing patterns are irregular (tachypnoea, Cheyne–Stokes); and eye movements are compromised – pupils are dilated and may be fixed. A state of decortication may convert to decerebrate rigidity, when all extremities are extended.

Lesions located in the temporal lobe may expand medially, forcing the uncus and possibly the parahippocampal gyrus over the edge of the tentorium cerebelli with impingement on the midbrain (see Fig. 21.20D). This is uncal herniation and may present in two forms. First, the midbrain injury involves structures on the side of the herniation with predictable deficits: ipsilateral pupil dilation and ophthalmoplegia, contralateral weakness of upper and lower extremities, an altered level of consciousness and hyperactive reflexes. Second, the herniation may shift the midbrain to the side opposite the herniation (e.g. with herniation on the right, the midbrain shifts to the left). In this case the dilated pupil and third nerve ophthalmoplegia are on the right, and the crus cerebri damaged on the left results in a right hemiplegia; both motor deficits are on the right side. The corticospinal deficit, in this case, is a false localizing sign.

Pressure increase in the posterior fossa, an infratentorial location, may result in herniation upwards through the tentorial notch or downwards into the foramen magnum. Upward cerebellar herniation (see Fig. 21.20E) displaces the brainstem against the tentorium and into the notch with resultant symptoms of increased intracranial pressure, paralysis of upward gaze, motor deficits and altered consciousness. Extrusion of the cerebellar tonsils (see Fig. 21.20F) downwards into the foramen magnum may result in damage to medullary cardiac and respiratory centres, with resultant changes in heart and respiratory rates. Both of these syndromes may constitute a medical emergency, the former due to rapidly developing hydrocephalus, the latter due to compromise of cardiac and respiratory centres.

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